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Supplementary material for: Neural architecture search via trie-based deduplication and zero-cost proxies, with hardware-aware coverage: An empirical study of LLM-guided warm-start

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Contents. This document gives the material that supports the main paper but did not fit within its page limit: the detailed search loop and its settings (section S1); the status of the formal statements and the landscape measurements behind them (section S2); full proxy and cost tables (section S3); every per-cell result at a fixed proposal budget (section S4) and at equal GPU-hours (section S5); all rank-based statistics (section S6); the proposer × gate grid (section S7); the ablations and sensitivity studies (section S8); NAS-Bench-101 (section S9); the LLM studies (section S10); hardware-aware search (section S11); and the seeds, reproducibility checks and data checksums (section S12). Supplementary objects are numbered S1, S2, ... and always named (Table S5, Figure S3); numbers without the prefix refer to the main paper, and S1–S4 written alone denote the development stages of the main paper. Every value in the tables and figures is computed by two scripts (`make_supp_tables.py`, `make_supp_figures.py`) directly from the result files in the public repository (https://github.com/Imaadhajwane/Arch-Tree-LLM); nothing is typed by hand. Test accuracies are in %, differences in percentage points (pt), and all results use 12 seeds unless stated otherwise. A difference is called a win or a loss only if it is significant after Holm correction within its family (paired two-sided Wilcoxon test, $p_{\text{Holm}} < 0.05$); W/T/L denotes wins/ties/losses of ArchTree-RMG (or of ArchTree where the gate is held fixed).

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Table S1. Identifiers in the result files and the corresponding names in the paper.

Identifier(s) in the CSV files	Name in the paper	Description
`archtree_v1`, `archtree_trie_only`	S1, ArchTree-base	trie, raw-reward UCB ($C = \sqrt{2}$, mean backup), no gate
`archtree_zc`, `archtree_phase2`	S2	S1 with the supervised gate (1,500 labelled architectures)
`archtree` + gate `none`	S3, ArchTree-RM	rank-normalised UCB ($C = 0.3$, max backup) with mutation, no gate
`archtree_llm`, `archtree_headline`, `archtree_v2`, `archtree_zc_unsup`, `v2`	S4, ArchTree-RMG	ArchTree-RM with the label-free gate; LLM prior off ($\alpha = 0$)
`archtree_llm_hw`	ArchTree-HW	rank UCB with max backup, no gate, no mutation, latency-penalised reward
`costpen`, `costpen025`	cost penalty $\lambda = 1, 0.25$	gate score $S_\lambda = z(P) - \lambda z(D)$, $D = \log(1 + \#\text{params})$
`zc`, `zc_unsup`, `params`	supervised, label-free, #params gate	gate applied to any proposer
`flat_ucb_none`, `flat_ucb_set`	flat UCB (A7)	per-position UCB without / with a set-based deduplication
`v1`, `rank`, `mut_only`, `re` (design)	design-ablation arms	base + gate; rank reward only; mutation only; RE with the same gate
`llm_off`, `llm_on`, `llm_described`, `size_prior`	LLM-study arms	$\alpha = 0$; LLM prior; operations described without names; FLOP-size prior without an LLM

Table S2. Settings of the search as read by the code (identical in every reported run unless an experiment varies them).

Component	Setting	Value	Varied in
UCB	exploration constant C	0.3 (S1: $\sqrt{2}$)	Table S28
	reward / backup	percentile rank / max (S1–S2: raw / mean)	Table S24
Mutation	share of proposals, start	0.5, after 10 real evaluations	Table S24
	tournament size	5 evaluated architectures	–
Gate	warm-up / pass rate	first 10 real evaluations / top 35% of scores seen	Figure 4
	proxies, calibration	E, S, T; z-scores from 1,500 unlabelled architectures	Table S29
	proxy network	32 channels, fixed initialisation seed 1234	–
	NASWOT batches	3 batches of 256 CIFAR-10 images	–
Pruning	period / share / eligibility	every 50 iterations / 15% / ≥ 2 visits, by value	Table S27
LLM prior	α, decay, floor	0 (headline), $\alpha/\sqrt{t'}$, stop below 0.02	Table S33
	model, temperature, max tokens	`openai/gpt-oss-20b`, 0.1, 300	Table S37
	calls per run	at most 60	–
Hardware	latency penalty	$0.05 \cdot \min(\text{lat}/50\,\text{ms}, 1)$ on the 0–1 reward	–
Protocol	search / report metric	200-epoch validation / test accuracy	–
	budgets	NB201: 25–200 proposals; NB101: 50–400	–

S1 Naming, search loop and settings

Table S1 maps the identifiers used in the result files to the names used in the paper, so that every number can be traced back to its source. Figure S1 expands the five-step summary of the search loop (Figure 2) into its seven operations, and Table S2 lists every setting that the code actually reads; keys present in the configuration file but not read by any script are not reported.

S2 Formal statements and the landscape behind them

The early design documents of the project made seven formal claims (T1–T7). Table S3 records what can and cannot be said for each, so that no statement in the main paper rests on an unproven result. The main paper uses T1, T2 and T5 only as properties of the design and makes no regret claim.

Prefix locality. Statement T3 needs architectures that share a prefix to have similar accuracy. Table S4 and Figure S2 measure this on the whole NAS-Bench-201 space. The first three edges explain little of the variance ($\eta^2 \leq 0.13$, 0.19 and 0.24 on the three datasets), and the spread within a three-token prefix (7.8–10.8 pt) is almost that of the whole space. Locality appears only at depth four ($\eta^2 = 0.47$–0.59). Statistics collected near the root of the trie therefore carry little information about unvisited leaves, which is consistent with the finding that the tree adds nothing beyond deduplication (Table S25). The number of depth-k cells that contain a leaf within 1 pt of the optimum grows as $5^{\gamma k}$ with $\gamma = 0.54$, 0.30 and 0.32; CIFAR-10 has many near-optimal architectures (376 leaves within 1 pt), which explains why all good methods saturate on it.

Regret. Figure S3 shows the simple regret of every method against real evaluations at the largest budget, and Table S5 the fitted power law $r_t \approx a\, t^{-b}$. Uniform sampling without replacement has $b = 0.29$–0.46, and random, MCTS-style, ENAS-style, ZiCo-style search and LLM-NAS stay below 0.7. ArchTree-RMG decays with $b = 1.74$, 1.71 and 1.29 and reaches a final regret of 0.04, 0.08 and 0.15 pt after 88–102 real evaluations; only local search decays clearly faster on CIFAR-10 and CIFAR-100 ($b = 2.68$ and 3.24), where it reaches the optimum. The A7 trie arm reproduces ArchTree-RMG exactly. The flat bandit with set deduplication has a similar exponent on CIFAR-10 (1.78) and a smaller one on CIFAR-100 and ImageNet16-

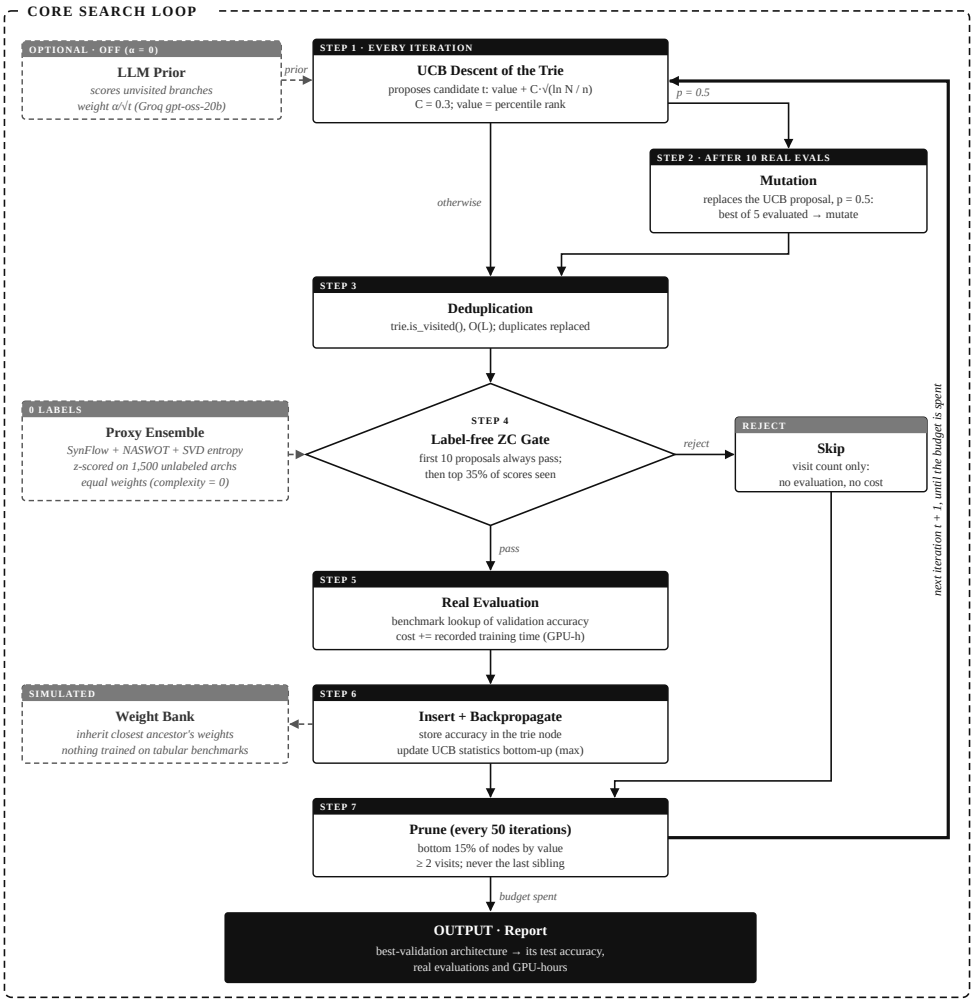


Figure S1. The search loop of ArchTree-RMG in detail. Each iteration descends the trie by rank-normalised UCB (step 1) or, with probability 0.5 after ten real evaluations, replaces the proposal by a tournament mutation (step 2); the trie rejects architectures already seen (step 3); the label-free gate admits every proposal until ten real evaluations have been made and then only those in the top 35% of the scores observed so far (step 4); admitted architectures receive a real evaluation (step 5), are inserted with their accuracy, which is backed up along the path (step 6), and every 50 iterations the lowest 15% of nodes by backed-up value are pruned (step 7). Dashed boxes are optional or zero-cost components: the LLM prior (off in the headline system), the proxy ensemble and the weight bank, which has no effect on tabular benchmarks.

Table S3. Status of the formal statements.

Id	Statement	Status	Comment
T1	trie insertion and look-up cost $O(L)$	holds	one step per token; $L = 6$ on NB201
T2	no architecture is evaluated twice	holds by construction	visited leaves are redirected to an unvisited one; confirmed empirically (0 repeats in every ArchTree run, Table S11)
T3	sub-linear regret of UCB over the trie	informal only	relies on prefix locality (Table S4); the max backup and rank reward depart from the settings of the UCT and HOO analyses [1, 2, 3]; treated as motivation, not as a guarantee
T4	pruning never removes the optimum	false as stated	value-based pruning is a heuristic without confidence bounds; its effect is measured instead (Table S27)
T5	weight inheritance is consistent	holds	no measurable effect on tabular benchmarks, where nothing is trained (A3: identical runs, Table S26)
T6	the LLM prior preserves the regret behaviour	near-trivial	the prior only reorders never-visited siblings; a zero-information prior reproduces $\alpha = 0$ exactly (24/24 seeds)
T7	hypervolume ratio equals 1 iff the whole Pareto front is found	definitional	used only as a descriptive metric

Table S4. Prefix locality of NAS-Bench-201 (all 15,625 architectures, validation accuracy): variance explained by the first k tokens (η^2), mean within-prefix standard deviation, and number of prefixes $N_k(\varepsilon)$ that contain an architecture within ε pt of the best; γ is the fitted growth exponent $N_k \propto 5^{\gamma k}$.

Dataset	k	prefixes	η^2	within s.d.	$N_k(0.25)$	$N_k(0.5)$	$N_k(1)$	$N_k(2)$	$N_k(4)$
CIFAR-10	1	5	0.063	11.55	3	4	5	5	5
	2	25	0.118	11.01	5	9	13	21	25
	3	125	0.131	10.77	10	22	46	90	125
	4	625	0.471	6.01	10	23	76	269	546
	5	3,125	0.577	4.10	17	51	212	811	2308
	6	15,625	1.000	0.00	21	70	376	1660	7190
				fitted γ	0.24	0.35	0.54	0.73	0.91
CIFAR-100	1	5	0.097	10.98	1	1	2	2	5
	2	25	0.165	10.42	1	2	4	8	20
	3	125	0.188	10.15	1	4	12	21	79
	4	625	0.495	6.53	1	4	12	23	250
	5	3,125	0.612	4.76	2	5	19	55	707
	6	15,625	1.000	0.00	2	5	22	84	1319
				fitted γ	0.10	0.19	0.30	0.44	0.71
ImageNet16-120	1	5	0.142	8.38	2	2	3	5	5
	2	25	0.219	7.98	3	6	9	16	20
	3	125	0.242	7.83	4	7	17	45	86
	4	625	0.587	5.17	5	9	22	88	261
	5	3,125	0.705	3.98	6	13	36	197	794
	6	15,625	1.000	0.00	6	17	46	278	1601
				fitted γ	0.14	0.24	0.32	0.50	0.73

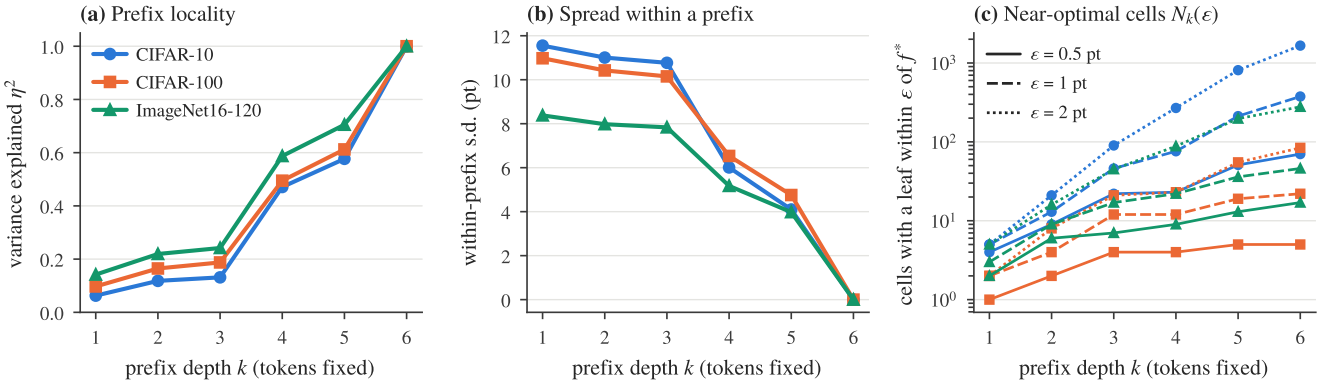


Figure S2. Landscape of NAS-Bench-201 seen through the trie. (a) Variance of validation accuracy explained by the first k tokens. (b) Mean standard deviation of the architectures that share a prefix of length k. (c) Number of prefixes that still contain an architecture within ε of the optimum (log scale). The first three tokens carry little information; locality appears from depth four.

120 (1.08, 0.90), although its final accuracy does not differ (Table S25); the flat bandit without deduplication is slower still (1.12, 0.66, 0.74). The exponents are descriptive summaries over 25–200 evaluations, not asymptotic rates.

S3 Proxies, training cost and the gate-cost criterion

Table S6 gives the rank correlation of each proxy with validation accuracy on the 1,500 calibration architectures. On NAS-Bench-201 the label-free ensemble P is within 0.04 of the supervised ensemble in Kendall τ (0.63–0.68 against 0.63–0.72), and NASWOT is the best single proxy. On NAS-Bench-101 all proxies are weaker (best single $\tau = 0.39$).

Table S7 extends Figure 6(b) to every score and space. Two observations matter for the gate. First, in the cell spaces the label-free score, SynFlow and NASWOT are also cost proxies: the label-free score correlates with training time with $\rho = 0.78$ (NB201) and 0.63 (NB101), so the admitted 35% train 1.22 and 1.35 times longer than an average architecture. Measured against an oracle that admits the truly best 35%, this over-payment is small on NAS-Bench-201 (3.4–5.1%) but reaches 18.0% on NAS-Bench-101. Second, in the width space of NATS-Bench SSS the correlation with cost is weak ($\rho = 0.20$–0.21) and

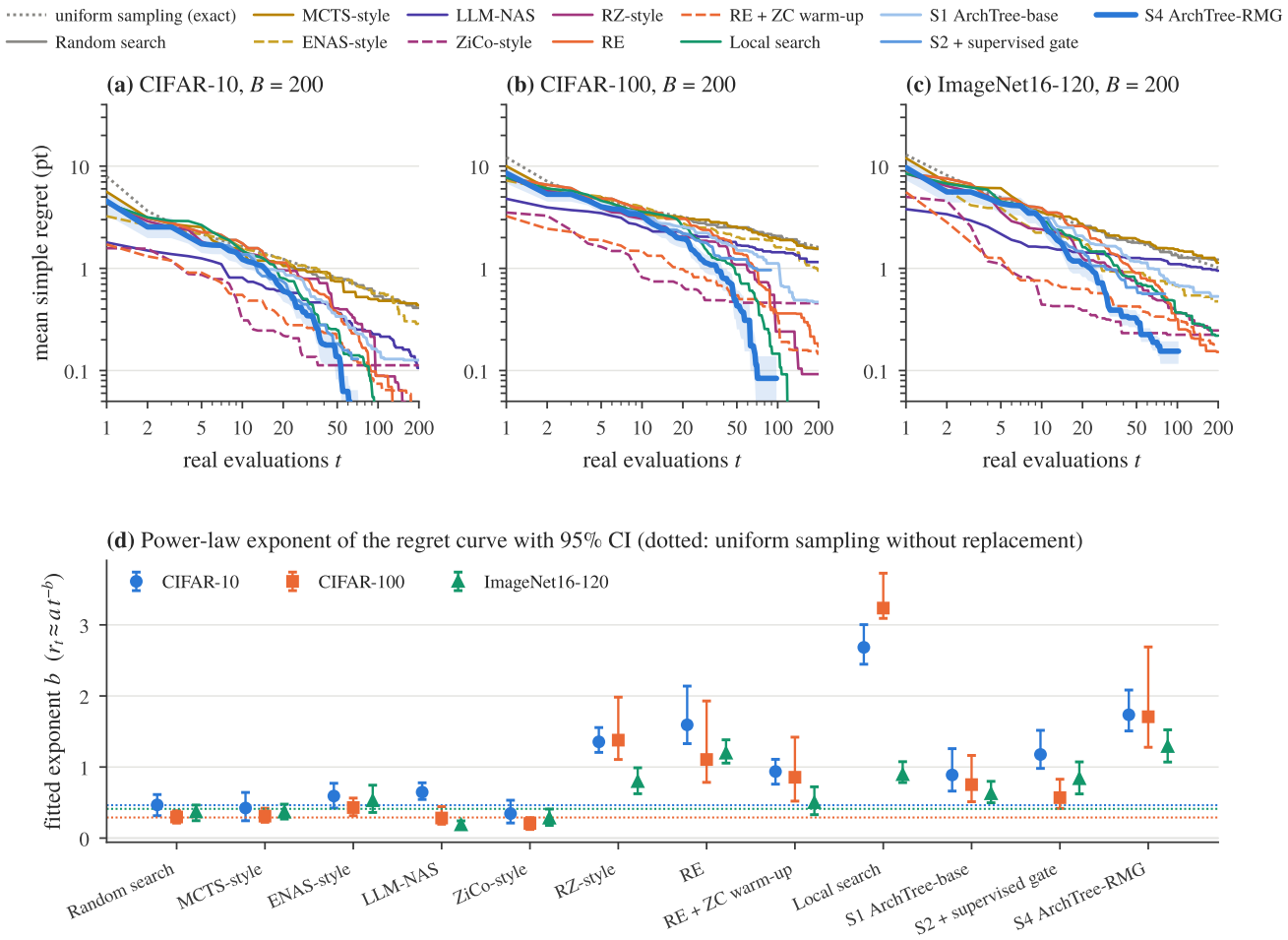


Figure S3. Gap between the optimum of the space and the best validation accuracy found so far (simple regret), NAS-Bench-201, 200 proposals. (a)–(c) Mean of 12 seeds against real evaluations (log–log; band: ±1 standard error for ArchTree-RMG). Gated methods stop earlier because they spend fewer real evaluations. (d) Fitted exponent b with 95% bootstrap interval per method and dataset; dotted lines mark uniform sampling without replacement.

Table S5. Power-law fit of the regret curve at a budget of 200 proposals: real evaluations T (maximum over seeds), exponent b with 95% bootstrap interval, and simple regret r_T (pt) at T.

Method	CIFAR-10			CIFAR-100			ImageNet16-120		
	T	b [95% CI]	r_T	T	b [95% CI]	r_T	T	b [95% CI]	r_T
Uniform sampling (exact)	200	0.46	0.42	200	0.29	1.60	200	0.41	1.02
Random search	200	0.47 [0.31, 0.61]	0.41	200	0.30 [0.21, 0.39]	1.55	200	0.37 [0.24, 0.46]	1.22
MCTS-style	200	0.42 [0.24, 0.64]	0.44	200	0.31 [0.22, 0.42]	1.55	200	0.37 [0.27, 0.48]	1.13
ENAS-style	200	0.59 [0.42, 0.77]	0.28	200	0.43 [0.31, 0.56]	0.95	200	0.53 [0.36, 0.74]	0.48
LLM-NAS	200	0.65 [0.54, 0.78]	0.11	200	0.28 [0.20, 0.44]	1.15	200	0.19 [0.14, 0.24]	0.95
ZiCo-style	200	0.34 [0.21, 0.53]	0.11	200	0.21 [0.12, 0.29]	0.46	200	0.28 [0.18, 0.41]	0.22
RZ-style ZC walk	200	1.35 [1.21, 1.56]	0.04	200	1.38 [1.11, 1.98]	0.09	200	0.79 [0.62, 0.99]	0.25
RE	200	1.59 [1.33, 2.14]	0.02	200	1.11 [0.78, 1.93]	0.17	200	1.20 [1.06, 1.38]	0.15
RE + ZC warm-up	200	0.94 [0.76, 1.11]	0.03	200	0.86 [0.52, 1.42]	0.15	200	0.50 [0.33, 0.72]	0.18
Local search	200	2.68 [2.45, 3.00]	0.00	200	3.24 [3.09, 3.73]	0.00	200	0.89 [0.78, 1.08]	0.22
S1 ArchTree-base	200	0.89 [0.66, 1.26]	0.13	200	0.75 [0.51, 1.16]	0.47	200	0.62 [0.50, 0.80]	0.53
S2 (S1 + supervised gate)	71	1.18 [0.98, 1.52]	0.13	89	0.57 [0.42, 0.83]	0.96	81	0.84 [0.62, 1.07]	0.56
S4 ArchTree-RMG	88	1.74 [1.51, 2.08]	0.04	98	1.71 [1.28, 2.69]	0.08	102	1.29 [1.07, 1.52]	0.15
Flat UCB, no dedup (A7)	142	1.12 [0.71, 1.76]	0.11	148	0.66 [0.37, 1.31]	0.66	144	0.74 [0.43, 1.17]	0.55
Flat UCB + set dedup (A7)	95	1.78 [1.47, 2.17]	0.02	98	1.08 [0.96, 1.25]	0.22	97	0.90 [0.72, 1.09]	0.38
Trie (A7 arm)	88	1.74 [1.51, 2.18]	0.04	98	1.71 [1.27, 3.67]	0.08	102	1.29 [1.03, 1.58]	0.15

Table S6. Rank correlation of each zero-cost proxy with validation accuracy (Kendall τ, Spearman ρ; 1,500 architectures, calibration seed 0). C is a cost proxy and enters only the cost-penalised gates.

Proxy	NB201 C10		NB201 C100		NB201 IN16		NB101 C10	
	τ	ρ	τ	ρ	τ	ρ	τ	ρ
E (SVD entropy)	0.55	0.73	0.54	0.72	0.49	0.66	0.39	0.55
S (SynFlow)	0.55	0.75	0.59	0.78	0.56	0.74	0.26	0.38
T (NASWOT)	0.59	0.79	0.64	0.83	0.59	0.78	0.27	0.39
C (FLOPs/params cost)	−0.50	−0.69	−0.53	−0.72	−0.48	−0.66	−0.37	−0.54
supervised ensemble	0.67	0.85	0.72	0.89	0.63	0.81	0.50	0.69
label-free ensemble P	0.64	0.83	0.68	0.86	0.63	0.81	–	–

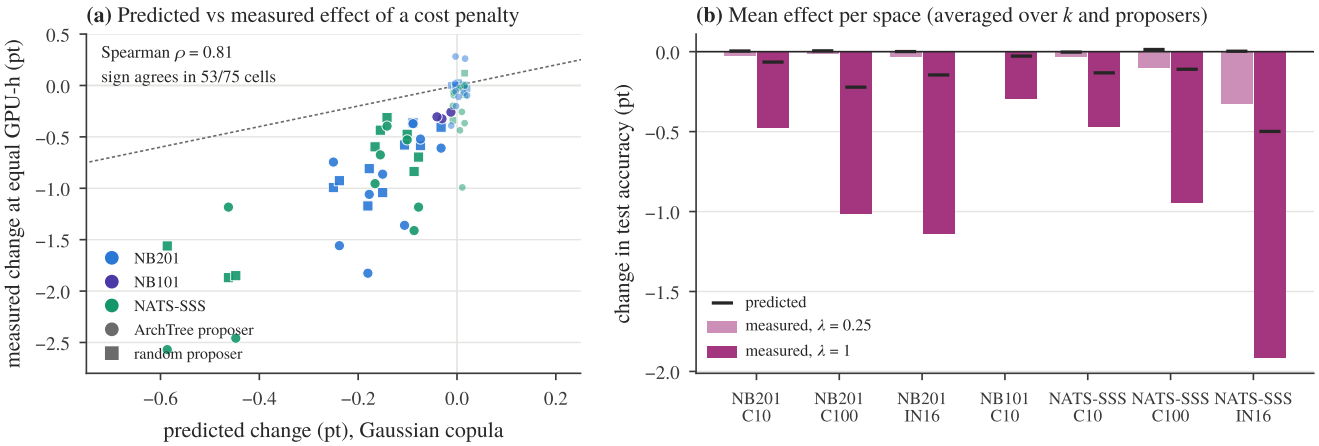


Figure S4. Validation of the gate-cost criterion. (a) Predicted against measured change in test accuracy when a cost penalty is added to the gate, for every benchmark, dataset, budget k, proposer and λ (large markers: $\lambda = 1$; small: $\lambda = 0.25$); the dotted line is $y = x$. (b) Mean measured change per space (bars) and the corresponding prediction (black ticks). Every space predicts and shows a loss.

the admitted set is cheaper than the oracle set (−1.7 to −6.4%). The parameter count, often used as a free gate, is tied to cost as strongly as or more strongly than any accuracy proxy (ρ = 0.68–0.87).

Gate-cost criterion. Table S8 gives all six normal-score correlations behind Eq. 7 together with the predicted and measured effect of a cost penalty $\lambda = 1$, and Figure S4 compares prediction and measurement for each of the 75 measured cells. The criterion is negative in all seven spaces and the measured effect is negative in all seven; across cells the predicted and measured changes are rank-correlated (ρ = 0.81 over both λ values, 0.71 for $\lambda = 1$ alone) and agree in sign in 53 of 75 cells, including all 39 cells with $\lambda = 1$; all 22 disagreements are $\lambda = 0.25$ cells in which the predicted change lies within ±0.02 pt although the measured loss reaches up to 0.99 pt. The Gaussian-copula model underestimates the size of the loss by a factor of about 4 to 10, because the gate acts on the upper tail of the score distribution, where the dependence between proxy and accuracy is stronger than a single correlation describes. The criterion should therefore be read as a test of direction, not as a forecast of magnitude.

S4 Fixed proposal budget: all cells

Table S9 gives every method on every dataset and budget; Table 5 of the main paper is its last column. Table S10 and Figure S5 summarise the 132 paired comparisons of ArchTree-RMG with each baseline. Two definitions of equal effort are used: the same number of proposals, which favours ungated methods since they evaluate every proposal, and a budget at which the baseline receives at least as many real evaluations as ArchTree-RMG used. The second definition changes the counts only slightly (21 wins and no loss against the external baselines). Against the nine external baselines ArchTree-RMG has 20 wins and no loss in 108 comparisons: 7 against random search, 6 against MCTS-style search, 3 each against ENAS-style search and LLM-NAS, and 1 against ZiCo-style search. Against RE with a ZC warm-up and local search the mean difference is slightly negative (−0.22 and −0.08 pt) but never significant.

Repeated evaluations. Table S11 and Figure S6 count real evaluations of architectures that the same run had already trained. All ArchTree variants have none (statement T2), as do the population and sampling baselines that keep a history. The

Table S7. Correlation of every gate score with training time τ (Spearman ρ) and with validation accuracy A (Kendall τ_K) over the space (NB201: all 15,625 architectures; NB101 and NATS-SSS: 2,000 random architectures), mean training time of the top 35% relative to the space average, excess of that cost over the oracle top 35%, and mean validation accuracy of the admitted architectures.

Space	Score	ρ(score, τ)	τ_K(score, A)	cost ratio of top 35%	excess vs oracle (%)	mean A of top 35%
NB201 C10	*E*	0.44	0.55	1.13	−3.1	88.54
	S	0.75	0.55	1.21	4.0	88.70
	T	0.82	0.59	1.23	6.3	88.83
	C	−0.88	−0.49	0.76	−34.8	77.99
	P (label-free)	0.78	0.64	1.22	5.1	88.90
	#params	0.83	0.53	1.21	4.6	87.80
	val. acc. (oracle)	0.59	1.00	1.16	0.0	89.39
NB201 C100	*E*	0.44	0.54	1.13	−4.7	67.15
	S	0.75	0.58	1.21	2.3	67.67
	T	0.82	0.63	1.23	4.6	67.92
	C	−0.88	−0.51	0.76	−35.9	54.40
	P (label-free)	0.78	0.67	1.22	3.4	68.00
	#params	0.83	0.55	1.21	2.9	66.62
	val. acc. (oracle)	0.65	1.00	1.18	0.0	68.67
NB201 IN16	*E*	0.44	0.49	1.13	−3.3	39.04
	S	0.75	0.56	1.21	3.8	39.79
	T	0.82	0.59	1.23	6.1	40.15
	C	−0.88	−0.47	0.76	−34.9	27.76
	P (label-free)	0.78	0.64	1.22	4.9	40.18
	#params	0.83	0.51	1.21	4.4	38.69
	val. acc. (oracle)	0.60	1.00	1.16	0.0	41.56
NB101 C10	*E*	0.10	0.38	1.05	−8.2	91.70
	S	0.63	0.23	1.36	18.5	91.34
	T	0.82	0.23	1.48	29.0	90.86
	C	−0.42	−0.38	0.80	−29.8	88.05
	P (label-free)	0.63	0.33	1.35	18.0	91.46
	#params	0.87	0.29	1.48	29.2	91.23
	val. acc. (oracle)	0.33	1.00	1.14	0.0	92.88
NATS-SSS C10	*E*	0.13	0.34	1.03	−7.5	88.62
	S	0.27	0.63	1.05	−5.2	89.00
	T	0.11	0.35	1.02	−8.0	88.58
	C	−0.25	−0.57	0.95	−14.6	86.92
	P (label-free)	0.20	0.53	1.04	−6.4	88.89
	#params	0.74	0.69	1.15	3.0	89.04
	val. acc. (oracle)	0.59	1.00	1.11	0.0	89.17
NATS-SSS C100	*E*	0.13	0.56	1.03	−3.7	66.32
	S	0.27	0.79	1.06	−1.1	67.20
	T	0.11	0.24	1.02	−4.3	64.49
	C	−0.25	−0.65	0.95	−11.4	58.17
	P (label-free)	0.20	0.63	1.04	−2.4	66.80
	#params	0.74	0.54	1.15	7.9	66.28
	val. acc. (oracle)	0.29	1.00	1.07	0.0	67.41
NATS-SSS IN16	*E*	0.14	0.49	1.01	−2.2	40.32
	S	0.27	0.74	1.02	−1.3	41.46
	T	0.14	0.29	1.01	−2.2	39.26
	C	−0.27	−0.58	0.98	−5.0	33.61
	P (label-free)	0.21	0.62	1.02	−1.7	41.09
	#params	0.68	0.65	1.05	1.3	41.14
	val. acc. (oracle)	0.46	1.00	1.03	0.0	41.80

Table S8. Gate-cost model per space: normal-score correlations between gate score P, accuracy A, predicted cost $D = \log(1 + \#\text{params})$ and recorded training time C; criterion slope $r_{PA}r_{PD} - r_{DA}$ at $\lambda = 0$; predicted and measured change (pt) of test accuracy at equal GPU-hours for $\lambda = 1$ (mean over k and proposers).

Space	r_{PA}	r_{PD}	r_{DA}	r_{PC}	r_{AC}	r_{DC}	slope	pred. Δ	meas. Δ
NB201 C10	0.83	0.79	0.69	0.77	0.59	0.85	−0.037	−0.07	−0.48
NB201 C100	0.86	0.79	0.72	0.77	0.65	0.85	−0.043	−0.22	−1.02
NB201 IN16	0.82	0.79	0.66	0.77	0.61	0.85	−0.019	−0.15	−1.14
NB101 C10	0.47	0.82	0.44	0.65	0.35	0.89	−0.049	−0.03	−0.30
NATS-SSS C10	0.74	0.67	0.89	0.23	0.62	0.76	−0.390	−0.13	−0.47
NATS-SSS C100	0.84	0.67	0.75	0.23	0.33	0.77	−0.185	−0.11	−0.95
NATS-SSS IN16	0.83	0.67	0.85	0.21	0.46	0.66	−0.293	−0.50	−1.92

Table S9. NAS-Bench-201: test accuracy (mean ± standard deviation over 12 seeds) of the best-validation architecture at each proposal budget B. Best value per column in bold.

Method	$B = 25$	$B = 50$	$B = 100$	$B = 200$
CIFAR-10				
Random search	93.45 ± 0.45	93.62 ± 0.21	93.82 ± 0.30	93.92 ± 0.21
MCTS-style	93.55 ± 0.36	93.68 ± 0.27	93.89 ± 0.15	93.89 ± 0.20
ENAS-style	93.43 ± 0.71	93.68 ± 0.48	93.74 ± 0.41	94.09 ± 0.26
LLM-NAS	93.93 ± 0.21	93.99 ± 0.25	94.04 ± 0.19	94.28 ± 0.08
ZiCo-style	**93.93 ± 0.22**	94.08 ± 0.20	94.17 ± 0.18	94.29 ± 0.12
RZ-style ZC walk	93.72 ± 0.27	94.06 ± 0.21	**94.28 ± 0.11**	94.33 ± 0.07
RE	93.42 ± 0.41	93.94 ± 0.36	94.17 ± 0.24	94.31 ± 0.13
RE + ZC warm-up	93.91 ± 0.33	**94.26 ± 0.15**	94.22 ± 0.19	94.28 ± 0.19
Local search	93.75 ± 0.43	93.97 ± 0.38	94.27 ± 0.22	**94.37 ± 0.00**
S1 ArchTree-base	93.64 ± 0.43	93.99 ± 0.19	94.09 ± 0.24	94.11 ± 0.26
S2 (S1 + supervised gate)	93.60 ± 0.44	93.98 ± 0.34	94.15 ± 0.25	94.17 ± 0.22
S4 ArchTree-RMG	93.60 ± 0.31	94.04 ± 0.27	94.19 ± 0.26	94.28 ± 0.13
CIFAR-100				
Random search	70.84 ± 0.64	71.06 ± 0.78	71.54 ± 0.83	71.92 ± 0.81
MCTS-style	70.59 ± 0.98	70.97 ± 0.91	71.62 ± 0.69	71.94 ± 0.87
ENAS-style	71.02 ± 1.25	71.58 ± 1.14	71.79 ± 1.15	72.59 ± 0.81
LLM-NAS	71.36 ± 0.51	71.39 ± 0.42	71.96 ± 0.28	72.14 ± 0.43
ZiCo-style	71.93 ± 0.80	72.33 ± 0.66	72.68 ± 0.58	72.97 ± 0.43
RZ-style ZC walk	71.39 ± 0.77	72.17 ± 0.99	73.20 ± 0.35	73.39 ± 0.19
RE	70.43 ± 1.22	71.61 ± 1.21	72.96 ± 0.78	73.37 ± 0.21
RE + ZC warm-up	**72.00 ± 1.03**	**72.92 ± 0.56**	73.13 ± 0.50	73.36 ± 0.23
Local search	71.44 ± 1.48	72.63 ± 1.09	**73.36 ± 0.42**	**73.51 ± 0.00**
S1 ArchTree-base	71.12 ± 1.35	72.06 ± 0.93	72.50 ± 0.85	73.12 ± 0.45
S2 (S1 + supervised gate)	70.98 ± 1.40	71.46 ± 1.37	72.26 ± 1.06	72.45 ± 0.89
S4 ArchTree-RMG	71.51 ± 1.26	72.19 ± 0.91	73.02 ± 0.75	73.40 ± 0.20
ImageNet16-120				
Random search	44.74 ± 1.25	45.15 ± 0.73	45.49 ± 0.85	45.47 ± 0.80
MCTS-style	44.24 ± 1.03	44.66 ± 0.99	45.14 ± 0.86	45.44 ± 0.65
ENAS-style	45.26 ± 1.12	45.84 ± 0.78	46.04 ± 0.44	46.16 ± 0.36
LLM-NAS	44.77 ± 0.34	45.35 ± 0.54	45.29 ± 0.30	45.37 ± 0.36
ZiCo-style	45.37 ± 1.04	46.11 ± 0.48	46.22 ± 0.42	46.28 ± 0.57
RZ-style ZC walk	45.53 ± 0.89	46.06 ± 0.46	46.22 ± 0.45	46.32 ± 0.18
RE	44.53 ± 1.35	45.67 ± 1.10	**46.51 ± 0.29**	46.48 ± 0.17
RE + ZC warm-up	**46.11 ± 0.57**	**46.31 ± 0.53**	46.14 ± 0.57	46.34 ± 0.36
Local search	45.23 ± 0.96	46.05 ± 0.79	46.11 ± 0.65	**46.52 ± 0.29**
S1 ArchTree-base	44.93 ± 1.08	45.70 ± 0.58	46.11 ± 0.46	46.25 ± 0.40
S2 (S1 + supervised gate)	44.90 ± 1.23	45.72 ± 1.08	45.81 ± 0.93	46.39 ± 0.51
S4 ArchTree-RMG	45.28 ± 1.07	45.91 ± 0.51	46.45 ± 0.39	46.41 ± 0.18

Table S10. Paired comparison of ArchTree-RMG with each baseline over the 12 dataset–budget cells: Holm-corrected W/T/L (Holm within baseline), range and mean of the paired difference Δ (pt, ArchTree-RMG minus baseline) and median Cohen's *d*; on the right, the same comparison when each baseline is given at least as many real evaluations as ArchTree-RMG used.

Baseline	Same proposal budget (12 cells)				Matched real evaluations (12 cells)	
	W/T/L	range of Δ	mean Δ	median *d*	W/T/L	mean Δ
Random search	7/5/0	0.14 to 1.48	0.77	1.01	7/5/0	0.81
MCTS-style	6/6/0	0.05 to 1.47	0.89	1.05	6/6/0	0.92
ENAS-style	3/9/0	0.01 to 1.23	0.42	0.60	4/8/0	0.52
LLM-NAS	3/9/0	−0.33 to 1.27	0.53	0.57	3/9/0	0.57
ZiCo-style	1/11/0	−0.42 to 0.43	−0.01	−0.04	1/11/0	0.03
RZ-style ZC walk	0/12/0	−0.25 to 0.23	−0.03	−0.11	0/12/0	−0.01
RE	0/12/0	−0.07 to 1.08	0.24	0.15	0/12/0	0.29
RE + ZC warm-up	0/12/0	−0.83 to 0.31	−0.22	−0.22	0/12/0	−0.20
Local search	0/12/0	−0.44 to 0.35	−0.08	−0.24	0/12/0	−0.06
S1 ArchTree-base	0/12/0	−0.04 to 0.52	0.22	0.37	1/11/0	0.27
S2 (S1 + supervised gate)	1/11/0	0.00 to 0.96	0.37	0.39	1/11/0	0.38

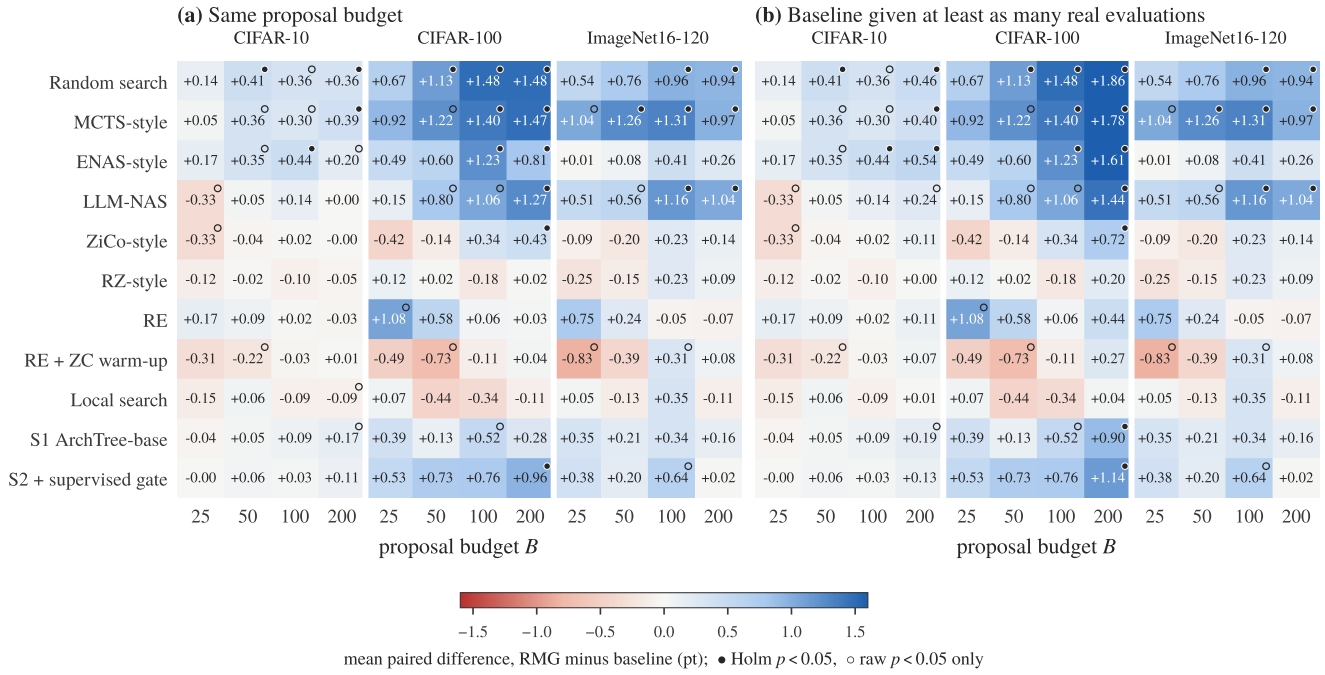


Figure S5. Mean paired difference in test accuracy between ArchTree-RMG and each baseline on NAS-Bench-201 (blue: ArchTree-RMG better). Filled dots: significant after Holm correction within the baseline; open dots: $p < 0.05$ before correction only. (a) Same number of proposals. (b) Baseline budget raised until it receives at least as many real evaluations as ArchTree-RMG.

LLM generator repeats itself increasingly as the budget grows (6.4% at 25 proposals, 55.4% at 200, up to 133 repeats in one run), and a flat bandit that keeps no memory wastes 60–75% of its evaluations on architectures it has already trained, which is the waste that deduplication removes.

S5 Equal GPU-hours: all cells

Tables S12–S14 list the mean test accuracy of every arm in every cell of the equal-GPU-hour protocol (Eq. 8), with a marker where ArchTree-RMG is significantly better. Table S15 counts the outcomes per comparator and benchmark, and Figure S7 shows every paired difference. ArchTree-RMG has 57 wins and no loss in 261 comparisons (38/79/0 on NAS-Bench-201, 4/23/0 on NAS-Bench-101 and 15/102/0 on NATS-Bench SSS). Most wins are against ungated random search (10) and the

Table S11. Repeated real evaluations: mean share over runs (%) and maximum number in a single run, per proposal budget B (NAS-Bench-201, mean over the three datasets).

Method	$B = 25$		$B = 50$		$B = 100$		$B = 200$	
	%	max	%	max	%	max	%	max
Random search	0.0	0	0.0	0	0.0	0	0.0	0
MCTS-style	0.2	1	0.2	1	0.2	1	0.3	3
ENAS-style	0.0	0	0.0	0	0.0	0	0.0	0
LLM-NAS	6.4	5	20.3	18	36.9	49	55.4	133
ZiCo-style	0.0	0	0.0	0	0.0	0	0.0	0
RZ-style ZC walk	0.0	0	0.0	0	0.0	0	0.0	0
RE	0.0	0	0.0	0	0.0	0	0.0	0
RE + ZC warm-up	0.0	0	0.0	0	0.0	0	0.0	0
Local search	0.0	0	0.0	0	0.0	0	0.0	0
S1 ArchTree-base	0.0	0	0.0	0	0.0	0	0.0	0
S2 (S1 + supervised gate)	0.0	0	0.0	0	0.0	0	0.0	0
S4 ArchTree-RMG	0.0	0	0.0	0	0.0	0	0.0	0
Flat UCB, no dedup (A7)	60.4	14	64.4	28	68.9	56	74.5	109
Flat UCB + set dedup (A7)	0.0	0	0.0	0	0.0	0	0.0	0
Trie (A7 arm)	0.0	0	0.0	0	0.0	0	0.0	0

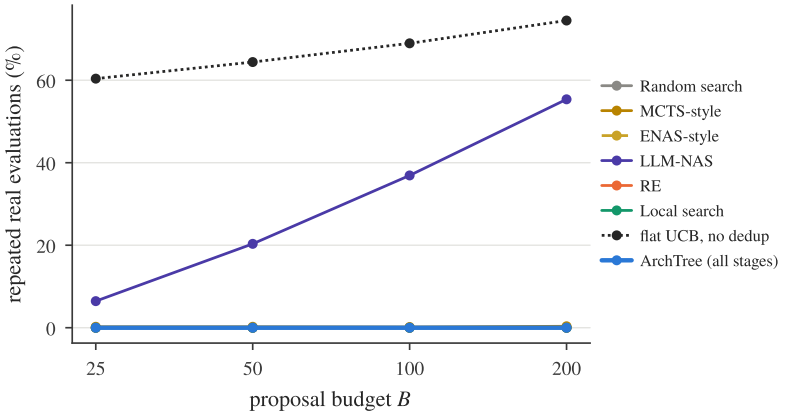


Figure S6. Share of real evaluations spent on architectures already evaluated in the same run (NAS-Bench-201, mean over datasets and seeds). Methods not shown, and all ArchTree stages, stay at or below 0.3%.

two $\lambda = 1$ cost penalties (31); on NAS-Bench-201 they also include random search with the label-free gate or with $\lambda = 0.25$ (6 each), the ungated ArchTree-RM (2), ungated RE (1) and the parameter-count gate (1). Against the other gated arms, RE with the ZC warm-up and local search, the mean differences lie between −0.24 and +0.41 pt, with a single significant cell (the parameter-count gate on NAS-Bench-201). Five of the 1,128 runs ran out of proposals before their GPU-hour budget was spent (all ArchTree-RM with $\lambda = 1$ on NAS-Bench-101 at $k = 100$, seeds 111, 789, 999, 1001 and 2024); they are kept with the accuracy they reached.

S6 Rank-based statistics

Table S16 reports the Friedman test for each of the five comparison sets, Tables S17 and S18 all average ranks, Tables S19 and S20 the probability that ArchTree-RMG finds a better architecture on a paired run, and Table S21 the equivalence tests. Equal performance of all methods is rejected everywhere ($p \leq 2.2 \times 10^{-8}$). ArchTree-RMG has the best average rank on NAS-Bench-201 at equal GPU-hours and lies inside the critical difference of the top method in the other four sets. The per-run probabilities separate the field more finely than the ranks: in paired runs ArchTree-RMG comes out ahead 75–85% of the time against MCTS-style, random and ENAS-style search and LLM-NAS, 0.50–0.57 against RE, ZiCo-style and RZ-style search, and 41–43% against RE with ZC warm-up and local search; at equal GPU-hours local search with the gate wins more paired runs on NAS-Bench-101 (0.22) and NATS-Bench SSS (0.38), as is RE with ZC warm-up on NATS-Bench SSS (0.40). Equivalence within ±0.5 pt is shown in at most 5 of 12 NAS-Bench-201 cells for any baseline and within ±0.25 pt in at most 3, so the ties in the Wilcoxon tests should be read as “no detectable difference”, not as equality.

Table S12. NAS-Bench-201 at equal GPU-hours: mean test accuracy for budget $H_k = k\bar{\tau}$. Bold: best arm in the column; $^{\triangledown}$: ArchTree-RMG significantly better ($p_{\text{Holm}} < 0.05$). No arm is significantly better than ArchTree-RMG in any cell.

Arm	CIFAR-10			CIFAR-100			ImageNet16-120		
	25	50	100	25	50	100	25	50	100
ArchTree-RMG	93.83	94.11	94.28	72.09	72.60	73.40	45.45	**46.49**	46.41
ArchTree-RM (no gate)	93.64	93.79$^{\triangledown}$	94.09	71.28	72.07	72.84	45.10	45.56$^{\triangledown}$	45.91
ArchTree-RM + #params gate	93.81	94.15	94.25	71.82	72.40	73.21	45.44	46.01$^{\triangledown}$	46.32
ArchTree-RM + cost pen. (λ=0.25)	93.75	94.13	94.17	71.99	72.88	73.20	45.71	46.10	46.35
ArchTree-RM + cost pen. (λ=1)	93.45$^{\triangledown}$	93.59$^{\triangledown}$	93.67$^{\triangledown}$	71.03	71.85	71.85$^{\triangledown}$	44.58	44.67$^{\triangledown}$	45.05$^{\triangledown}$
RE + gate	93.67	94.16	94.35	71.43	72.71	73.28	44.97	45.79	46.39
RE + ZC warm-up	**94.10**	**94.20**	94.24	**72.56**	72.88	73.05	**45.88**	45.95	46.16
RE	93.55	93.90	94.08	70.77$^{\triangledown}$	71.48	72.76	45.06	46.04	46.24
LS + gate	93.78	94.08	**94.37**	71.29	**72.90**	**73.51**	45.04	45.61	**46.47**
LS	93.68	93.94	94.19	70.77	72.09	73.14	45.17	45.75	46.25
Random + gate	93.62	93.83	93.92$^{\triangledown}$	71.06$^{\triangledown}$	71.56$^{\triangledown}$	71.92$^{\triangledown}$	45.09	45.49$^{\triangledown}$	45.45$^{\triangledown}$
Random	93.45	93.62$^{\triangledown}$	93.82$^{\triangledown}$	70.84$^{\triangledown}$	71.06$^{\triangledown}$	71.54$^{\triangledown}$	44.71	45.15$^{\triangledown}$	45.53$^{\triangledown}$
Random + cost pen. (λ=0.25)	93.60	93.83	93.95$^{\triangledown}$	71.03$^{\triangledown}$	71.54$^{\triangledown}$	71.92$^{\triangledown}$	45.10	45.49$^{\triangledown}$	45.45$^{\triangledown}$
Random + cost pen. (λ=1)	93.26$^{\triangledown}$	93.25$^{\triangledown}$	93.52$^{\triangledown}$	70.25$^{\triangledown}$	70.56$^{\triangledown}$	71.00$^{\triangledown}$	44.05$^{\triangledown}$	44.32$^{\triangledown}$	44.87$^{\triangledown}$

Table S13. NATS-Bench SSS at equal GPU-hours (same layout as Table S12).

Arm	CIFAR-10			CIFAR-100			ImageNet16-120		
	25	50	100	25	50	100	25	50	100
ArchTree-RMG	92.95	93.24	93.18	69.58	69.83	70.23	44.46	45.60	45.62
ArchTree-RM (no gate)	92.65	92.88	93.05	69.10	69.46	69.82	44.04	44.99	45.36
ArchTree-RM + #params gate	93.00	93.07	93.25	69.64	69.69	70.19	44.78	45.23	45.46
ArchTree-RM + cost pen. (λ=0.25)	92.85	93.17	93.15	69.33	69.83	69.86	44.26	44.61	45.18
ArchTree-RM + cost pen. (λ=1)	92.55$^{\triangledown}$	92.56$^{\triangledown}$	92.65$^{\triangledown}$	68.63$^{\triangledown}$	68.65$^{\triangledown}$	68.82$^{\triangledown}$	43.28$^{\triangledown}$	43.03$^{\triangledown}$	43.16$^{\triangledown}$
RE + gate	93.02	93.14	**93.36**	69.48	69.76	**70.34**	44.75	45.26	45.51
RE + ZC warm-up	93.14	93.18	93.18	69.64	69.86	70.13	45.39	**45.87**	45.69
RE	92.91	93.03	93.25	69.00	69.70	69.94	44.17	44.91	45.46
LS + gate	**93.17**	**93.29**	93.32	**69.76**	**70.13**	70.16	**45.41**	45.84	**45.75**
LS	93.09	93.20	93.34	69.74	69.95	70.14	44.67	45.76	45.50
Random + gate	92.97	93.10	93.10	69.26	69.40	69.69	44.90	44.81	45.26
Random	92.99	92.97	93.08	69.17	69.43	69.33	44.40	44.74	44.71
Random + cost pen. (λ=0.25)	92.97	93.07	93.12	69.26	69.31	69.81	44.56	44.81	45.26
Random + cost pen. (λ=1)	92.65	92.66$^{\triangledown}$	92.63$^{\triangledown}$	68.66	68.71$^{\triangledown}$	68.86$^{\triangledown}$	43.03	43.24$^{\triangledown}$	43.41$^{\triangledown}$

Table S14. NAS-Bench-101 (CIFAR-10) at equal GPU-hours: mean ± standard deviation of test accuracy and mean number of real evaluations that fit into each budget. Markers as in Table S12.

Arm	k = 25	k = 50	k = 100	real evals (k = 25 / 50 / 100)
ArchTree-RMG	**93.63 ± 0.20**	93.71 ± 0.18	93.82 ± 0.18	27 / 48 / 88
ArchTree-RM (no gate)	93.49 ± 0.26	93.56 ± 0.21	93.78 ± 0.24	32 / 60 / 116
ArchTree-RM + #params gate	93.47 ± 0.21	93.60 ± 0.14	93.93 ± 0.22	26 / 46 / 85
ArchTree-RM + cost pen. (λ=1)	93.37 ± 0.26	93.39 ± 0.23	93.51 ± 0.25$^{\triangledown}$	35 / 73 / 143
RE + gate	93.47 ± 0.25	93.72 ± 0.22	**93.96 ± 0.24**	22 / 42 / 80
RE + ZC warm-up	93.46 ± 0.21	93.56 ± 0.21	93.89 ± 0.20	11 / 28 / 76
RE	93.36 ± 0.23	93.56 ± 0.18	93.84 ± 0.27	26 / 56 / 108
LS + gate	93.47 ± 0.78	**93.92 ± 0.28**	93.95 ± 0.27	20 / 34 / 62
LS	93.32 ± 0.84	93.66 ± 0.34	93.83 ± 0.24	23 / 44 / 87
Random	93.16 ± 0.38$^{\triangledown}$	93.34 ± 0.17$^{\triangledown}$	93.41 ± 0.20$^{\triangledown}$	25 / 49 / 100

Table S15. Equal GPU-hours: W/T/L of ArchTree-RMG against each arm (Holm correction within each comparator over all 21 cells of the three benchmarks) and mean paired difference (pt) per benchmark (12 seeds per cell; "–": arm not run on that benchmark).

Comparator	NB201 (9 cells)		NB101 (3 cells)		NATS-SSS (9 cells)	
	W/T/L	mean Δ	W/T/L	mean Δ	W/T/L	mean Δ
ArchTree-RM (no gate)	2/7/0	0.49	0/3/0	0.11	0/9/0	0.37
ArchTree-RM + #params gate	1/8/0	0.14	0/3/0	0.05	0/9/0	0.04
ArchTree-RM + cost pen. (λ=0.25)	0/9/0	0.04	–	–	0/9/0	0.27
ArchTree-RM + cost pen. (λ=1)	6/3/0	0.99	1/2/0	0.30	9/0/0	1.26
RE + gate	0/9/0	0.21	0/3/0	0.00	0/9/0	0.01
RE + ZC warm-up	0/9/0	−0.04	0/3/0	0.08	0/9/0	−0.16
RE	1/8/0	0.53	0/3/0	0.13	0/9/0	0.26
LS + gate	0/9/0	0.18	0/3/0	−0.06	0/9/0	−0.24
LS	0/9/0	0.41	0/3/0	0.12	0/9/0	−0.08
Random + gate	6/3/0	0.75	–	–	0/9/0	0.24
Random	7/2/0	0.99	3/0/0	0.42	0/9/0	0.43
Random + cost pen. (λ=0.25)	6/3/0	0.75	–	–	0/9/0	0.28
Random + cost pen. (λ=1)	9/0/0	1.51	–	–	6/3/0	1.20
All comparisons	38/79/0		4/23/0		15/102/0	

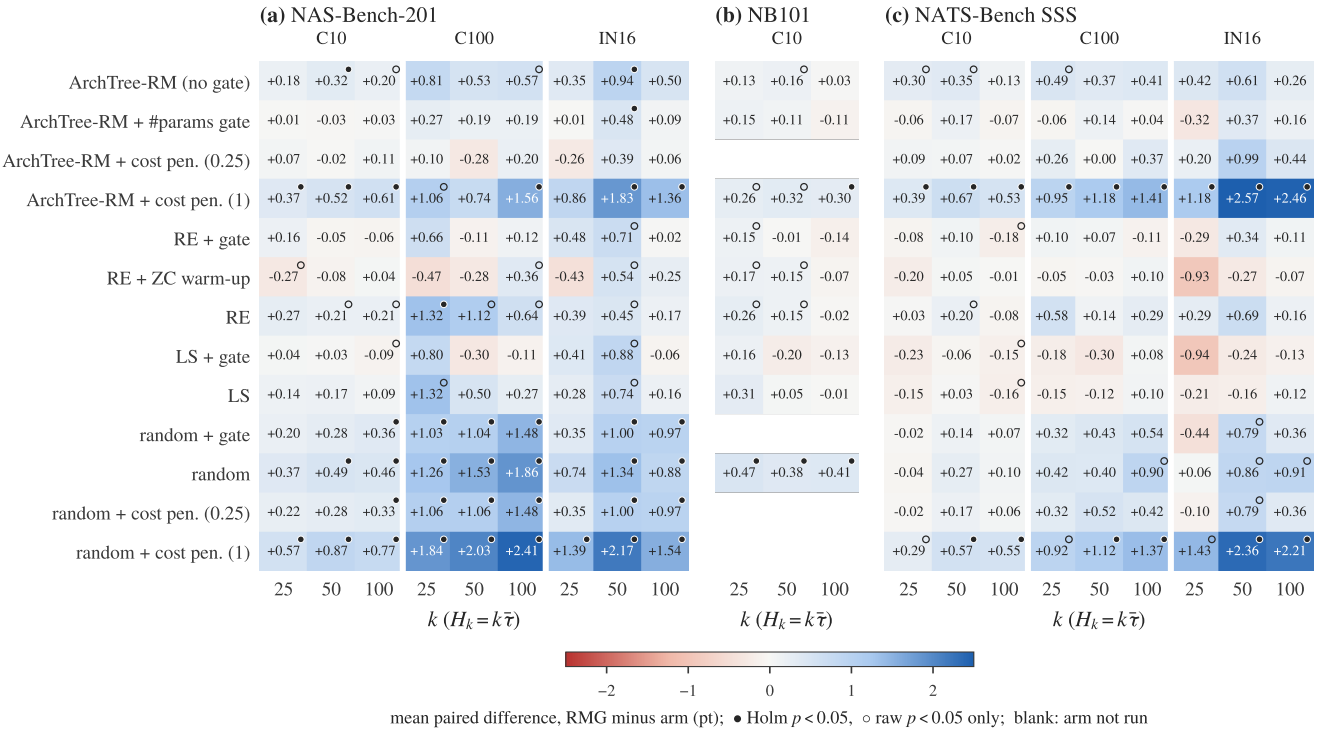


Figure S7. Mean paired difference in test accuracy between ArchTree-RMG and every other arm at equal GPU-hours (blue: ArchTree-RMG better), per dataset and budget multiple k. Filled dots: significant after Holm correction; open dots: $p < 0.05$ before correction only.

Table S16. Friedman test per comparison set (blocks = dataset × budget × seed) with the Nemenyi critical difference (CD), the average rank of ArchTree-RMG and the best method.

Comparison	methods	blocks	χ_F^2	p	CD	rank of RMG	best method (rank)
NB201, fixed proposal budget	10	144	381.2	1.4×10^{-76}	1.13	4.41	RE + ZC warm-up (3.67)
NB201, equal GPU-hours	14	108	428.1	2.5×10^{-83}	1.91	4.75	ArchTree-RMG (4.75)
NB101, fixed proposal budget	3	48	35.3	2.2×10^{-8}	0.48	1.73	RE (1.58)
NB101, equal GPU-hours	10	36	86.6	8.0×10^{-15}	2.26	4.38	LS + gate (3.11)
NATS-SSS, equal GPU-hours	14	108	416.9	6.0×10^{-81}	1.91	5.99	LS + gate (4.19)

Table S17. Average Friedman rank at a fixed proposal budget (1 = best). *: outside the CD of ArchTree-RMG.

Method	NB201, fixed proposal budget	NB101, fixed proposal budget
RE + ZC warm-up	3.67	–
Local search	3.90	–
ArchTree-RMG	**4.41**	**1.73**
RZ-style ZC walk	4.41	–
ZiCo-style	4.47	–
RE	5.07	1.58
ENAS-style	6.57*	–
LLM-NAS	6.70*	–
Random search	7.85*	–
MCTS-style	7.97*	–
Random	–	2.69*

Table S18. Average Friedman rank at equal GPU-hours (1 = best). *: outside the CD of ArchTree-RMG.

Method	NB201, equal GPU-hours	NB101, equal GPU-hours	NATS-SSS, equal GPU-hours
ArchTree-RMG	**4.75**	**4.38**	**5.99**
RE + ZC warm-up	4.84	5.11	5.45
ArchTree-RM + cost pen. (λ=0.25)	5.36	–	7.42
ArchTree-RM + #params gate	5.47	5.08	6.15
LS + gate	5.50	3.11	4.19
RE + gate	6.03	4.56	6.01
LS	6.91*	5.03	5.55
RE	8.04*	5.81	7.46
ArchTree-RM (no gate)	8.16*	5.88	8.74*
Random + gate	9.07*	–	7.80
Random + cost pen. (λ=0.25)	9.15*	–	7.94*
ArchTree-RM + cost pen. (λ=1)	9.58*	7.57*	11.93*
Random	10.20*	8.49*	8.66*
Random + cost pen. (λ=1)	11.94*	–	11.74*

Table S19. Probability that ArchTree-RMG finds a better architecture than the comparator on a paired (cell, seed) run at a fixed proposal budget, with stratified-bootstrap 95% interval. Bold: interval above 0.5; italic: interval below 0.5.

Comparator	NB201, fixed proposal budget	NB101, fixed proposal budget
RE + ZC warm-up	*0.41 [0.34, 0.48]*	–
Local search	*0.43 [0.36, 0.49]*	–
RZ-style ZC walk	0.50 [0.42, 0.57]	–
ZiCo-style	0.51 [0.44, 0.58]	–
RE	0.57 [0.50, 0.64]	0.44 [0.30, 0.57]
LLM-NAS	**0.75 [0.69, 0.81]**	–
ENAS-style	**0.77 [0.70, 0.83]**	–
Random search	**0.81 [0.75, 0.88]**	–
MCTS-style	**0.85 [0.79, 0.90]**	–
Random	–	**0.83 [0.73, 0.94]**
Paired runs	n = 144	n = 48

Table S20. Probability that ArchTree-RMG finds a better architecture at equal GPU-hours (layout as Table S19).

Comparator	NB201, equal GPU-hours	NB101, equal GPU-hours	NATS-SSS, equal GPU-hours
LS + gate	0.50 [0.42, 0.58]	*0.22 [0.08, 0.36]*	*0.38 [0.29, 0.46]*
RE + ZC warm-up	0.51 [0.43, 0.59]	0.58 [0.42, 0.72]	*0.40 [0.32, 0.49]*
ArchTree-RM + cost pen. (λ=0.25)	0.53 [0.45, 0.60]	–	0.57 [0.49, 0.65]
ArchTree-RM + #params gate	0.56 [0.49, 0.64]	0.58 [0.44, 0.72]	0.49 [0.41, 0.58]
RE + gate	**0.60 [0.52, 0.69]**	0.50 [0.35, 0.65]	0.51 [0.43, 0.60]
LS	**0.61 [0.52, 0.69]**	**0.67 [0.51, 0.81]**	0.47 [0.38, 0.56]
ArchTree-RM (no gate)	**0.76 [0.69, 0.83]**	**0.72 [0.58, 0.86]**	**0.69 [0.61, 0.76]**
RE	**0.79 [0.72, 0.86]**	0.62 [0.49, 0.75]	**0.62 [0.54, 0.72]**
ArchTree-RM + cost pen. (λ=1)	**0.85 [0.79, 0.90]**	**0.81 [0.67, 0.92]**	**0.89 [0.84, 0.94]**
Random + cost pen. (λ=0.25)	**0.85 [0.79, 0.91]**	–	**0.69 [0.60, 0.77]**
Random + gate	**0.85 [0.79, 0.91]**	–	**0.67 [0.58, 0.75]**
Random	**0.90 [0.84, 0.94]**	**0.92 [0.81, 1.00]**	**0.72 [0.64, 0.80]**
Random + cost pen. (λ=1)	**0.94 [0.89, 0.97]**	–	**0.91 [0.85, 0.95]**
Paired runs	n = 108	n = 36	n = 108

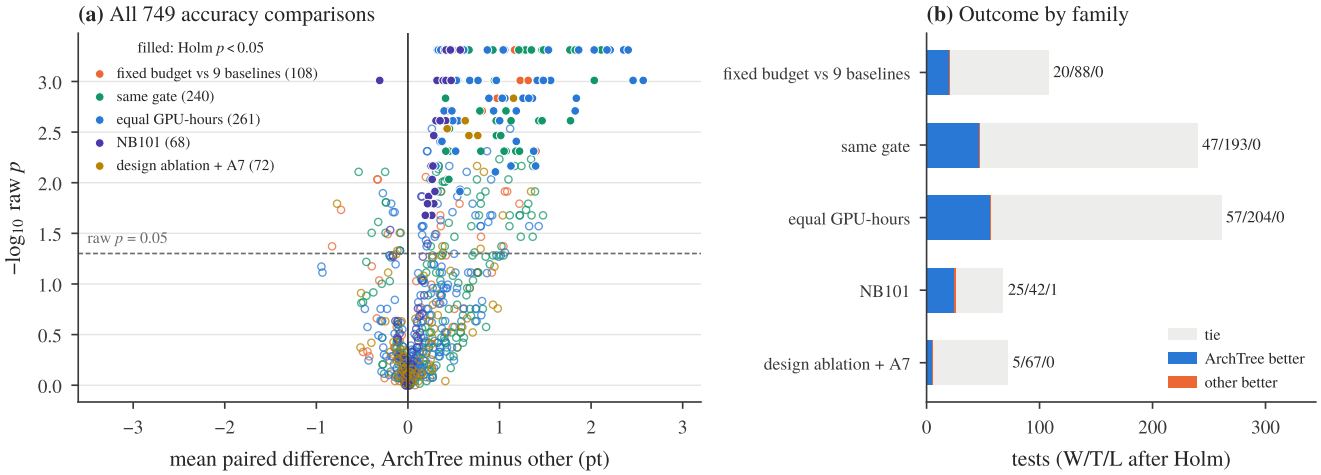


Figure S8. The 749 paired accuracy comparisons of ArchTree with another method or design variant in the five families of panel (b). (a) Mean paired difference against $-\log_{10}$ of the raw p-value; filled markers are significant after Holm correction (within baseline, arm or comparison, as in Table 8 of the main paper). The cap near $p \approx 5 \times 10^{-4}$ is the smallest p-value an exact Wilcoxon test can give with 12 pairs. (b) Holm-corrected W/T/L per family.

Figure S8 places the 749 accuracy comparisons of the main comparison families (listed in its panel b) on one plot; the sensitivity studies are excluded. The significant results lie almost entirely on the side favourable to ArchTree (154 of 155); the single loss is ArchTree-RM against RE, both with the parameter-count gate, on NAS-Bench-101 at a budget of 100 proposals.

S7 The gate on every proposer

Table S22 gives all 24 proposer–gate combinations at a budget of 200 proposals, Figure S9 the change in accuracy and the GPU-hours saved by each gate in every cell, and Table S23 the comparison of ArchTree-RM with each other proposer under the same gate. At a fixed number of proposals, the label-free gate saves 25–72% of GPU-hours across proposers, datasets and budgets. Its effect on accuracy depends on the proposer: it helps ArchTree-RM slightly (mean +0.11 pt) and leaves random search unchanged, but costs MCTS-style search 0.36 pt on average, because a proposer that generates few good candidates loses more when most of them are discarded. With the same gate, ArchTree-RM is never worse than any other proposer (W/T/L 7/5/0 against random, 9/3/0 against MCTS-style, 3/9/0 against ENAS-style search and 0/12/0 against RE and local search).

S8 Ablations and sensitivity

Held-out design ablation. Table S24 repeats the base→RMG design changes on 12 seeds that were not consulted for any design choice. Only four of 48 comparisons are significant, all in favour of ArchTree-RMG: the base system with the gate

Table S21. Two one-sided tests (TOST, Holm-corrected): number of cells in which ArchTree-RMG and the comparator are shown to be equivalent within ±0.5 and ±0.25 pt, and the largest absolute mean difference over the cells.

Comparison	Comparator	cells	equiv. at ±0.5	equiv. at ±0.25	max |Δ|
NB201, fixed proposal budget	ENAS-style	12	1	0	1.23
	LLM-NAS	12	3	1	1.27
	Local search	12	5	1	0.44
	MCTS-style	12	1	0	1.47
	Random search	12	0	0	1.48
	RE	12	4	2	1.08
	RE + ZC warm-up	12	5	1	0.83
	RZ-style ZC walk	12	5	3	0.25
	ZiCo-style	12	3	1	0.43
NB201, equal GPU-hours	ArchTree-RM + cost pen. (λ=1)	9	0	0	1.83
	ArchTree-RM + cost pen. (λ=0.25)	9	4	1	0.39
	ArchTree-RM (no gate)	9	2	0	0.94
	ArchTree-RM + #params gate	9	3	1	0.48
	LS	9	1	0	1.32
	LS + gate	9	4	1	0.88
	Random + cost pen. (λ=1)	9	0	0	2.41
	Random + cost pen. (λ=0.25)	9	1	0	1.48
	Random	9	0	0	1.86
	Random + gate	9	0	0	1.48
	RE	9	2	0	1.32
	RE + gate	9	4	2	0.71
	RE + ZC warm-up	9	2	1	0.54
NB101, fixed proposal budget	Random	4	2	0	0.41
	RE	4	4	1	0.14
NB101, equal GPU-hours	ArchTree-RM + cost pen. (λ=1)	3	1	0	0.32
	ArchTree-RM (no gate)	3	3	1	0.16
	ArchTree-RM + #params gate	3	3	0	0.15
	LS	3	2	1	0.31
	LS + gate	3	2	0	0.20
	Random	3	0	0	0.47
	RE	3	3	1	0.26
	RE + gate	3	3	1	0.15
	RE + ZC warm-up	3	3	0	0.17
NATS-SSS, equal GPU-hours	ArchTree-RM + cost pen. (λ=1)	9	0	0	2.57
	ArchTree-RM + cost pen. (λ=0.25)	9	3	1	0.99
	ArchTree-RM (no gate)	9	1	0	0.61
	ArchTree-RM + #params gate	9	3	0	0.37
	LS	9	2	1	0.21
	LS + gate	9	2	1	0.94
	Random + cost pen. (λ=1)	9	0	0	2.36
	Random + cost pen. (λ=0.25)	9	3	0	0.79
	Random	9	2	0	0.91
	Random + gate	9	3	0	0.79
	RE	9	3	0	0.69
	RE + gate	9	3	0	0.34
	RE + ZC warm-up	9	3	2	0.93

on all three datasets at 200 proposals, and the rank reward alone on ImageNet16-120. Mutation alone and RE with the same gate are statistically indistinguishable from ArchTree-RMG at every budget; at 200 proposals RE is within 0.11 pt of ArchTree-RMG on every dataset but uses 99–105 real evaluations against 84–90.

Trie against a flat bandit (A7). Table S25 compares the trie with a per-position UCB that either remembers evaluated architectures in a set or does not. With the set, no difference reaches $p < 0.05$ (12 cells), so the hierarchy itself adds nothing measurable. Without it, the flat bandit spends 11.7–94.7 evaluations per run on repeats and is worse in three cells before correction and one after (ImageNet16-120, $B = 100$).

Toggle ablations (A1–A6). Table S26 shows that removing the LLM prior (A1), weight inheritance (A3) or rerunning the budget grid (A6) reproduces the headline runs exactly (144 of 144 runs each, same accuracy and same number of evaluations), as expected with $\alpha = 0$ and tabular benchmarks; these rows document that the components are inert, not that they were tested for an effect. Removing the gate (A2) uses every proposal and changes the mean by −0.43 to +0.16 pt. Table S27 varies the share of pruned nodes: between 0 and 35% the mean accuracy moves by less than 0.2 pt, except a drop of 0.41 pt on

Table S22. Proposer × gate grid on NAS-Bench-201 at a budget of 200 proposals: test accuracy (mean ± standard deviation), mean real evaluations, and GPU-hours summed over the three datasets.

Proposer	Gate	CIFAR-10		CIFAR-100		ImageNet16-120		GPU-h
		acc.	evals	acc.	evals	acc.	evals	(3 sets)
ArchTree-RM	none	94.12 ± 0.24	200	72.97 ± 0.56	200	46.39 ± 0.45	200	1102
	#params	94.32 ± 0.14	92	73.27 ± 0.31	97	46.27 ± 0.33	87	570
	supervised	94.26 ± 0.21	76	73.27 ± 0.23	79	46.43 ± 0.33	81	514
	label-free	94.28 ± 0.13	81	73.40 ± 0.20	82	46.41 ± 0.18	83	520
RE	none	94.31 ± 0.13	200	73.37 ± 0.21	200	46.48 ± 0.17	200	1162
	#params	94.37 ± 0.00	116	73.34 ± 0.37	116	46.39 ± 0.51	112	716
	supervised	94.30 ± 0.14	93	73.51 ± 0.00	95	46.43 ± 0.18	106	649
	label-free	94.37 ± 0.00	105	73.44 ± 0.16	106	46.39 ± 0.20	103	652
LS	none	94.37 ± 0.00	200	73.51 ± 0.00	200	46.52 ± 0.29	200	1155
	#params	94.37 ± 0.00	90	73.51 ± 0.00	94	46.37 ± 0.65	100	631
	supervised	94.31 ± 0.21	84	73.51 ± 0.00	81	46.16 ± 0.81	87	559
	label-free	94.37 ± 0.00	79	73.51 ± 0.00	83	46.51 ± 0.70	90	564
Random	none	93.92 ± 0.21	200	71.92 ± 0.81	200	45.47 ± 0.80	200	925
	#params	93.90 ± 0.24	84	71.92 ± 0.82	84	45.48 ± 0.75	84	459
	supervised	93.92 ± 0.21	80	71.92 ± 0.81	79	45.47 ± 0.80	79	434
	label-free	93.92 ± 0.21	79	71.92 ± 0.81	79	45.47 ± 0.80	79	433
ENAS-style	none	94.09 ± 0.26	200	72.59 ± 0.81	200	46.16 ± 0.36	200	1068
	#params	94.01 ± 0.22	84	72.25 ± 1.07	86	45.78 ± 0.69	81	494
	supervised	93.81 ± 0.33	78	72.49 ± 0.84	75	45.71 ± 0.93	74	453
	label-free	93.96 ± 0.22	77	72.41 ± 1.06	78	45.86 ± 0.74	74	455
MCTS-style	none	93.88 ± 0.19	200	71.76 ± 0.86	200	45.60 ± 0.64	200	938
	#params	93.73 ± 0.32	67	71.43 ± 0.83	69	45.20 ± 0.90	76	380
	supervised	93.94 ± 0.18	61	71.45 ± 0.55	54	45.19 ± 0.73	52	272
	label-free	93.62 ± 0.42	56	71.29 ± 0.69	61	44.87 ± 0.82	54	276

Table S23. ArchTree-RM against each other proposer when both use the same gate: Holm-corrected W/T/L over 12 dataset–budget cells and mean paired difference (pt).

Other proposer	no gate		#params		supervised		label-free	
	W/T/L	mean Δ	W/T/L	mean Δ	W/T/L	mean Δ	W/T/L	mean Δ
Random	3/9/0	0.66	6/6/0	0.70	2/10/0	0.71	7/5/0	0.77
RE	0/12/0	0.13	0/12/0	0.12	1/11/0	0.20	0/12/0	0.20
LS	0/12/0	−0.19	0/12/0	−0.07	0/12/0	0.12	0/12/0	0.09
ENAS-style	0/12/0	0.31	1/11/0	0.49	3/9/0	0.51	3/9/0	0.53
MCTS-style	2/10/0	0.76	6/6/0	1.01	4/8/0	0.97	9/3/0	1.23

ImageNet16-120 at 35%, which is consistent with the moderate default of 15%.

UCB constant and backup. Table S28 and Figure S10 vary C from 0.1 to 2 under both backup rules. None of the 78 comparisons with the default is significant after correction (9 before). Large constants make the descent close to uniform over children (no duplicate proposals for $C \geq \sqrt{2}$), whereas small constants repeat good branches and lean more on deduplication; the max backup is the more robust rule, particularly on CIFAR-100, where the mean backup loses up to 0.75 pt.

Proxy subsets. Table S29 and Figure S11 run the search with every subset of the four proxies as gate. The ranking quality of a subset predicts search accuracy only on ImageNet16-120 (Spearman 0.54, $p = 0.04$; CIFAR-10 0.02, CIFAR-100 0.15). SVD entropy alone is the only subset that is significantly worse than the four-proxy combination (CIFAR-10 −0.45 pt and CIFAR-100 −1.11 pt; the only 2 of 48 comparisons significant after correction); all other subsets lie within the seed-to-seed spread of each other. In this ablation each subset is combined with weights fitted on the calibration architectures, so it isolates the choice of proxies; the EST subset reaches $\tau = 0.66$, 0.71 and 0.62, within 0.04 of the best subset on every dataset, and the label-free equal-weight version used in the paper ranks almost as well (Table S6).

S9 NAS-Bench-101

Table S30 gives every proposer–gate–budget cell on NAS-Bench-101, Table S31 the comparisons under the same gate and Table S32 the comparisons with ungated searches that receive at least as many real evaluations (11 wins, all against random

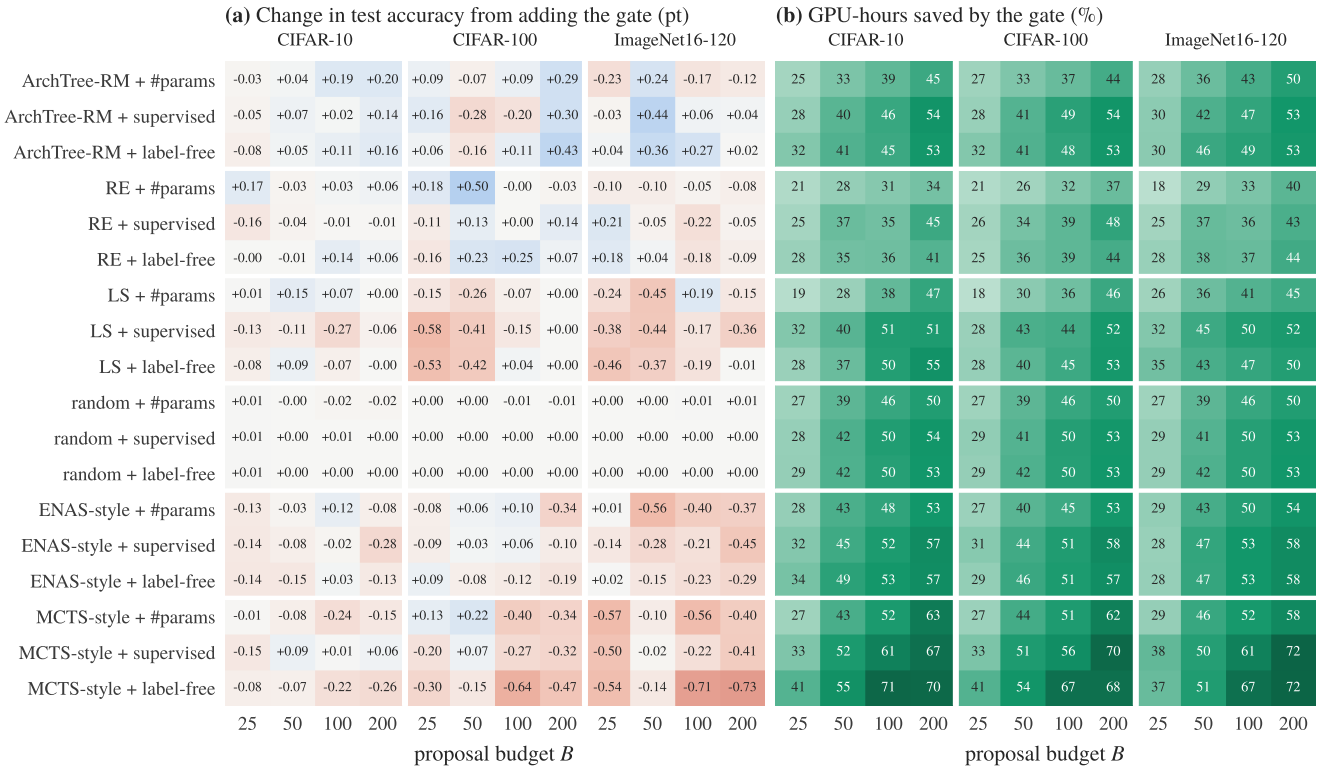


Figure S9. Effect of each gate on each proposer at a fixed number of proposals (NAS-Bench-201, 12 seeds). (a) Mean test accuracy with the gate minus without it. (b) Share of GPU-hours saved by the gate.

Table S24. Held-out design ablation (seeds 7–909): test accuracy (mean ± standard deviation) of each variant, all with the label-free gate, and real evaluations at $B = 200$. $^{\triangledown}$: ArchTree-RMG significantly better (Holm within arm over 12 cells, $p_{\text{Holm}} < 0.05$).

Arm	$B = 25$	$B = 50$	$B = 100$	$B = 200$	evals (B=200)
CIFAR-10					
ArchTree-base + gate	93.24 ± 0.60	93.46 ± 0.43	93.79 ± 0.38	93.83 ± 0.30$^{\triangledown}$	83.8
rank reward only	93.51 ± 0.56	93.88 ± 0.51	94.00 ± 0.28	94.17 ± 0.22	71.0
mutation only	93.62 ± 0.45	93.94 ± 0.37	94.23 ± 0.21	94.32 ± 0.14	93.7
RE + same gate	93.56 ± 0.53	94.03 ± 0.31	94.30 ± 0.18	94.37 ± 0.00	99.3
ArchTree-RMG	93.51 ± 0.44	93.81 ± 0.28	94.17 ± 0.23	94.26 ± 0.18	83.6
CIFAR-100					
ArchTree-base + gate	70.40 ± 1.20	70.93 ± 1.08	71.54 ± 1.05	72.27 ± 0.78$^{\triangledown}$	82.9
rank reward only	71.33 ± 1.06	71.98 ± 1.07	72.42 ± 0.96	72.59 ± 0.97	65.8
mutation only	71.09 ± 1.12	71.82 ± 1.17	72.98 ± 0.82	73.38 ± 0.27	92.9
RE + same gate	70.76 ± 1.09	72.06 ± 0.46	72.79 ± 0.92	73.37 ± 0.21	105.2
ArchTree-RMG	71.35 ± 0.86	71.97 ± 0.89	72.89 ± 0.76	73.42 ± 0.17	84.0
ImageNet16-120					
ArchTree-base + gate	44.17 ± 1.27	45.04 ± 1.12	45.35 ± 0.67	45.72 ± 0.50$^{\triangledown}$	82.8
rank reward only	44.63 ± 1.09	45.55 ± 0.68	45.39 ± 0.68	45.62 ± 0.60$^{\triangledown}$	71.0
mutation only	44.70 ± 0.92	45.31 ± 1.00	46.13 ± 0.63	46.14 ± 0.60	98.8
RE + same gate	45.11 ± 0.80	45.70 ± 0.82	46.39 ± 0.27	46.45 ± 0.19	101.5
ArchTree-RMG	44.59 ± 1.25	45.30 ± 0.92	45.62 ± 0.87	46.39 ± 0.27	90.4

Table S25. Trie against a flat per-position UCB with and without set-based deduplication: test accuracy (mean ± standard deviation), mean unique and repeated real evaluations, and paired difference (pt) of the trie with the unadjusted Wilcoxon p-value.

Dataset	B	trie		flat UCB + set dedup		flat UCB, no dedup		trie minus (p)	
		acc.	uniq./dup.	acc.	uniq./dup.	acc.	uniq./dup.	flat+set	flat
CIFAR-10	25	93.60 ± 0.31	16.4 / 0.0	93.76 ± 0.40	16.5 / 0.0	93.27 ± 0.59	7.9 / 11.7	−0.16 (0.375)	0.33 (0.203)
	50	94.04 ± 0.27	27.1 / 0.0	94.14 ± 0.30	26.9 / 0.0	93.83 ± 0.47	13.2 / 22.6	−0.10 (0.465)	0.20 (0.465)
	100	94.19 ± 0.26	47.8 / 0.0	94.22 ± 0.21	47.8 / 0.0	94.14 ± 0.35	21.8 / 45.7	−0.03 (0.676)	0.05 (0.910)
	200	94.28 ± 0.13	80.8 / 0.0	94.32 ± 0.13	83.9 / 0.0	94.22 ± 0.35	34.6 / 94.2	−0.04 (0.469)	0.06 (0.812)
CIFAR-100	25	71.51 ± 1.26	16.4 / 0.0	71.10 ± 1.17	17.2 / 0.0	70.53 ± 1.28	7.7 / 11.8	0.41 (0.910)	0.98 (0.175)
	50	72.19 ± 0.91	27.2 / 0.0	72.38 ± 1.10	26.8 / 0.0	71.48 ± 1.53	12.7 / 22.2	−0.19 (0.470)	0.71 (0.301)
	100	73.02 ± 0.75	47.1 / 0.0	72.76 ± 0.90	48.0 / 0.0	72.23 ± 1.14	21.5 / 46.6	0.26 (0.391)	0.79 (0.045)
	200	73.40 ± 0.20	81.5 / 0.0	73.14 ± 0.37	86.2 / 0.0	72.76 ± 1.15	32.8 / 94.1	0.26 (0.055)	0.65 (0.219)
ImageNet16-120	25	45.28 ± 1.07	16.5 / 0.0	45.13 ± 1.19	17.5 / 0.0	44.35 ± 1.67	7.8 / 11.8	0.15 (0.770)	0.93 (0.160)
	50	45.91 ± 0.51	24.8 / 0.0	45.80 ± 1.14	26.7 / 0.0	45.16 ± 0.78	12.6 / 24.1	0.12 (0.910)	0.76 (0.007)
	100	46.45 ± 0.39	45.4 / 0.0	46.33 ± 0.77	47.2 / 0.0	45.83 ± 0.63	21.4 / 48.2	0.13 (0.557)	0.63 (0.002)
	200	46.41 ± 0.18	82.8 / 0.0	46.45 ± 0.42	83.3 / 0.0	46.12 ± 0.65	33.0 / 94.7	−0.04 (0.676)	0.29 (0.276)

Table S26. Toggle ablations on NAS-Bench-201: mean test accuracy and, in brackets, mean real evaluations.

Dataset	B	S4 ArchTree-RMG	A1 no LLM	A2 no gate	A3 no inheritance	A6 budget grid
CIFAR-10	25	93.60 (16.4)	93.60 (16.4)	93.68 (25.0)	93.60 (16.4)	93.60 (16.4)
	50	94.04 (27.1)	94.04 (27.1)	93.99 (50.0)	94.04 (27.1)	94.04 (27.1)
	100	94.19 (47.8)	94.19 (47.8)	94.07 (100.0)	94.19 (47.8)	94.19 (47.8)
	200	94.28 (80.8)	94.28 (80.8)	94.12 (200.0)	94.28 (80.8)	94.28 (80.8)
CIFAR-100	25	71.51 (16.4)	71.51 (16.4)	71.45 (25.0)	71.51 (16.4)	71.51 (16.4)
	50	72.19 (27.2)	72.19 (27.2)	72.35 (50.0)	72.19 (27.2)	72.19 (27.2)
	100	73.02 (47.1)	73.02 (47.1)	72.91 (100.0)	73.02 (47.1)	73.02 (47.1)
	200	73.40 (81.5)	73.40 (81.5)	72.97 (200.0)	73.40 (81.5)	73.40 (81.5)
ImageNet16-120	25	45.28 (16.5)	45.28 (16.5)	45.24 (25.0)	45.28 (16.5)	45.28 (16.5)
	50	45.91 (24.8)	45.91 (24.8)	45.56 (50.0)	45.91 (24.8)	45.91 (24.8)
	100	46.45 (45.4)	46.45 (45.4)	46.18 (100.0)	46.45 (45.4)	46.45 (45.4)
	200	46.41 (82.8)	46.41 (82.8)	46.39 (200.0)	46.41 (82.8)	46.41 (82.8)

Table S27. Pruning share (A5) at $B = 100$: test accuracy (mean ± standard deviation) and mean real evaluations.

Pruned share	CIFAR-10	CIFAR-100	ImageNet16-120
0%	94.19 ± 0.26 (47.6)	73.16 ± 0.33 (48.1)	46.36 ± 0.50 (45.9)
15% (default)	94.19 ± 0.26 (47.8)	73.02 ± 0.75 (47.1)	46.45 ± 0.39 (45.4)
25%	94.22 ± 0.22 (47.5)	73.04 ± 0.59 (46.8)	46.34 ± 0.55 (46.6)
35%	94.22 ± 0.20 (47.2)	73.19 ± 0.36 (47.0)	46.04 ± 0.60 (46.4)

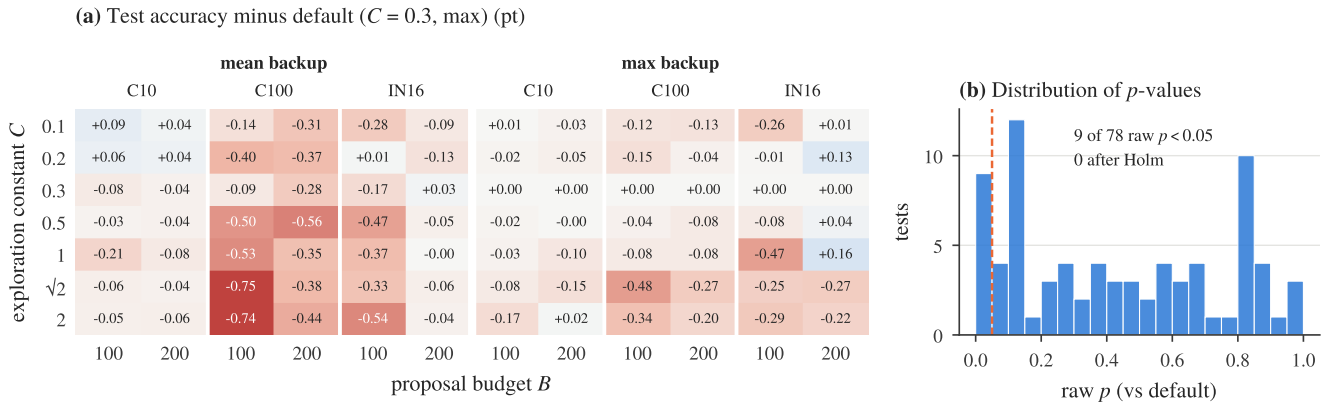


Figure S10. UCB sensitivity. (a) Test accuracy relative to the default ($C = 0.3$, max backup) for every C, backup rule, dataset and budget. (b) Distribution of the 78 unadjusted p-values against the default.

Table S28. UCB sensitivity: mean test accuracy for each exploration constant C and backup rule at $B = 100$ and 200, and mean number of proposals redirected because they were duplicates. Bold: default.

Backup	C	CIFAR-10		CIFAR-100		ImageNet16-120		dup.
		100	200	100	200	100	200	proposals
mean	0.1	94.28	94.32	72.88	73.09	46.17	46.32	29.6
	0.2	94.25	94.33	72.62	73.03	46.47	46.28	23.6
	0.3	94.10	94.24	72.93	73.13	46.28	46.44	14.7
	0.5	94.16	94.24	72.52	72.84	45.99	46.37	8.5
	1	93.98	94.20	72.49	73.06	46.09	46.41	0.3
	$\sqrt{2}$	94.13	94.24	72.27	73.02	46.13	46.36	0.0
	2	94.14	94.23	72.27	72.97	45.91	46.37	0.0
max	0.1	94.19	94.25	72.90	73.27	46.19	46.42	36.6
	0.2	94.16	94.24	72.87	73.36	46.45	46.54	27.7
	0.3	**94.19**	**94.28**	**73.02**	**73.40**	**46.45**	**46.41**	17.4
	0.5	94.16	94.28	72.98	73.33	46.37	46.45	10.7
	1	94.15	94.18	72.94	73.32	45.98	46.57	0.2
	$\sqrt{2}$	94.11	94.13	72.54	73.13	46.21	46.14	0.0
	2	94.02	94.30	72.68	73.21	46.16	46.19	0.0

Table S29. Proxy subsets as gate ($B = 100$): Kendall τ of the subset score with validation accuracy, test accuracy (mean ± standard deviation) and mean real evaluations. Subsets are combined with fitted weights.

Proxies	CIFAR-10			CIFAR-100			ImageNet16-120		
	τ	acc.	evals	τ	acc.	evals	τ	acc.	evals
E	0.54	93.65 ± 0.35	42.7	0.52	71.59 ± 1.18	44.6	0.48	45.83 ± 0.43	47.4
S	0.54	94.24 ± 0.20	48.2	0.57	73.04 ± 0.57	48.2	0.55	46.22 ± 0.56	47.4
T	0.59	94.19 ± 0.26	47.2	0.62	72.89 ± 0.71	46.9	0.58	46.30 ± 0.51	44.0
C	0.48	94.26 ± 0.20	50.5	0.49	72.64 ± 0.78	50.8	0.46	46.18 ± 0.59	47.6
ES	0.70	94.05 ± 0.22	43.5	0.70	72.66 ± 1.01	45.8	0.61	46.46 ± 0.40	46.2
ET	0.59	93.76 ± 0.32	42.0	0.62	71.95 ± 1.29	41.8	0.60	46.23 ± 0.44	45.9
EC	0.66	94.18 ± 0.28	45.2	0.65	72.74 ± 1.22	47.9	0.57	46.37 ± 0.57	44.7
ST	0.56	94.16 ± 0.21	47.8	0.59	72.86 ± 0.80	47.6	0.56	46.19 ± 0.66	47.2
SC	0.50	94.09 ± 0.26	47.8	0.56	72.92 ± 0.57	46.5	0.55	46.14 ± 0.58	46.8
TC	0.57	94.24 ± 0.18	46.8	0.57	72.64 ± 0.95	47.9	0.52	46.50 ± 0.46	46.1
EST	0.66	94.18 ± 0.35	44.5	0.71	72.90 ± 0.77	46.7	0.62	46.47 ± 0.39	47.1
ESC	0.69	94.27 ± 0.14	46.2	0.70	72.80 ± 0.85	46.7	0.62	46.27 ± 0.57	47.1
ETC	0.65	94.02 ± 0.35	44.9	0.67	72.69 ± 0.95	48.1	0.60	46.44 ± 0.68	44.2
STC	0.50	94.08 ± 0.28	46.9	0.58	73.06 ± 0.55	47.2	0.56	46.19 ± 0.66	47.2
ESTC	0.66	94.10 ± 0.30	46.1	0.70	72.70 ± 0.92	45.7	0.63	46.24 ± 0.63	46.8
no gate	–	94.07 ± 0.27	100.0	–	72.91 ± 0.59	100.0	–	46.18 ± 0.41	100.0

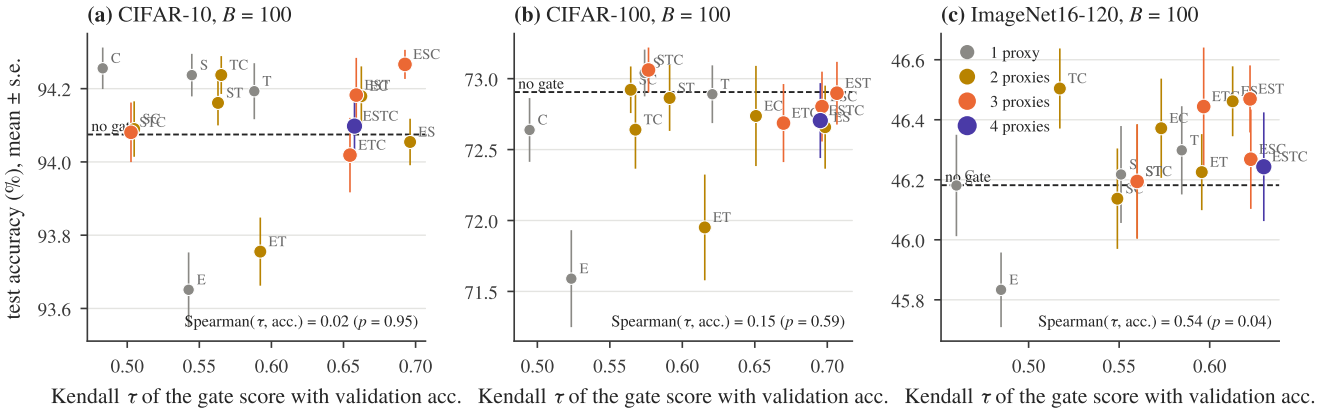


Figure S11. Ranking quality of each proxy subset against the accuracy reached with it as gate (mean ± standard error; colour and size: number of proxies). The dashed line is the search without a gate.

Table S30. NAS-Bench-101 (CIFAR-10): test accuracy (mean ± standard deviation) and, in brackets, mean real evaluations for each proposer, gate and proposal budget B.

Proposer	Gate	$B = 50$	$B = 100$	$B = 200$	$B = 400$
ArchTree-RM	none	93.51 ± 0.22 (50)	93.71 ± 0.23 (100)	93.81 ± 0.24 (200)	93.94 ± 0.20 (400)
	#params	93.51 ± 0.24 (27)	93.60 ± 0.14 (47)	93.98 ± 0.19 (85)	93.91 ± 0.21 (163)
	supervised	93.54 ± 0.30 (26)	93.69 ± 0.25 (45)	93.88 ± 0.24 (82)	93.96 ± 0.24 (158)
	label-free	93.60 ± 0.19 (28)	93.76 ± 0.18 (48)	93.82 ± 0.18 (89)	93.96 ± 0.19 (163)
RE	none	93.53 ± 0.21 (50)	93.76 ± 0.26 (100)	93.92 ± 0.25 (200)	94.13 ± 0.12 (400)
	#params	93.55 ± 0.24 (26)	93.90 ± 0.19 (48)	93.98 ± 0.18 (90)	94.07 ± 0.15 (164)
	supervised	93.50 ± 0.28 (25)	93.73 ± 0.21 (44)	93.98 ± 0.21 (84)	94.05 ± 0.19 (155)
	label-free	93.50 ± 0.23 (27)	93.78 ± 0.25 (44)	93.96 ± 0.24 (82)	94.07 ± 0.20 (154)
Random	none	93.38 ± 0.16 (50)	93.41 ± 0.20 (100)	93.49 ± 0.13 (200)	93.68 ± 0.12 (400)
	#params	93.33 ± 0.22 (27)	93.41 ± 0.20 (44)	93.49 ± 0.13 (79)	93.68 ± 0.12 (150)
	supervised	93.28 ± 0.26 (26)	93.39 ± 0.20 (43)	93.47 ± 0.18 (81)	93.68 ± 0.12 (150)
	label-free	93.34 ± 0.22 (28)	93.35 ± 0.23 (43)	93.44 ± 0.15 (78)	93.68 ± 0.12 (147)

Table S31. NAS-Bench-101: ArchTree-RM against random search and RE under the same gate (W/T/L over four budgets, mean paired difference in pt).

Gate (same for both)	ArchTree-RM vs random		ArchTree-RM vs RE	
	W/T/L	mean Δ	W/T/L	mean Δ
none	3/1/0	0.250	0/4/0	−0.096
#params	3/1/0	0.270	0/3/1	−0.127
supervised	4/0/0	0.308	0/4/0	−0.047
label-free	4/0/0	0.334	0/4/0	−0.044

Table S32. NAS-Bench-101 with matched evaluations: gated ArchTree-RM at proposal budget B against ungated random search, RE and ArchTree-RM given the smallest budget at least as large as the gated run's maximum number of real evaluations ("other B"). Mean paired difference in pt with p_{Holm} in brackets; bold: significant win.

Gate	B	max evals	other B	vs random	vs RE	vs ArchTree-RM
#params	50	32	50	0.13 (0.301)	−0.02 (1.000)	0.01 (1.000)
	100	53	100	**0.19 (0.042)**	−0.17 (0.160)	−0.12 (1.000)
	200	93	100	**0.57 (0.002)**	0.21 (0.160)	**0.26 (0.037)**
	400	182	200	**0.42 (0.002)**	−0.01 (1.000)	0.10 (1.000)
supervised	50	32	50	0.16 (0.092)	0.01 (1.000)	0.03 (1.000)
	100	52	100	**0.28 (0.032)**	−0.07 (1.000)	−0.02 (1.000)
	200	92	100	**0.47 (0.002)**	0.11 (1.000)	0.16 (0.161)
	400	171	200	**0.46 (0.002)**	0.03 (1.000)	0.15 (0.055)
label-free	50	31	50	**0.21 (0.016)**	0.06 (1.000)	0.09 (0.699)
	100	59	100	**0.35 (0.005)**	−0.01 (1.000)	0.04 (0.699)
	200	98	100	**0.42 (0.004)**	0.06 (1.000)	0.11 (0.699)
	400	177	200	**0.47 (0.004)**	0.04 (1.000)	0.15 (0.699)

search or the ungated ArchTree-RM); Figure S12 plots them. ArchTree-RM beats random search in 14 of 16 cells and ties RE in 15, the exception being a loss to RE with the parameter-count gate at 100 proposals. The gates leave 39–55% of the real evaluations of ArchTree-RM (for example 162.8 instead of 400 with the label-free gate) without a consistent loss of accuracy, although the gain is smaller than on NAS-Bench-201 because the proxies rank NAS-Bench-101 less well (Table S6).

S10 LLM prior

Prior strength. Table S33 and Figure S13(a,b) sweep α. For $\alpha \geq 0.2$ the runs are identical to those at $\alpha = 0.2$, because the decayed weight $\alpha/\sqrt{t'}$ falls below the floor of 0.02 at the same iteration for all of them. The largest effect is +0.85 pt on CIFAR-100 at 50 proposals, while ImageNet16-120 is negative in 7 of its 8 distinct (α, B) cells (down to −0.63 pt). Because the runs for $\alpha \geq 0.2$ coincide, the 48 comparisons with $\alpha = 0$ contain 24 distinct ones; none survives Holm correction (3 have $p < 0.05$ before it), so the prior was left off in the headline system.

Speed of convergence. Table S34 and Figure S14 measure how quickly each arm approaches the best validation accuracy in the space at $B = 100$. On CIFAR-10 the LLM prior needs 5.0 real evaluations to come within 1 pt (15.5 without it), and

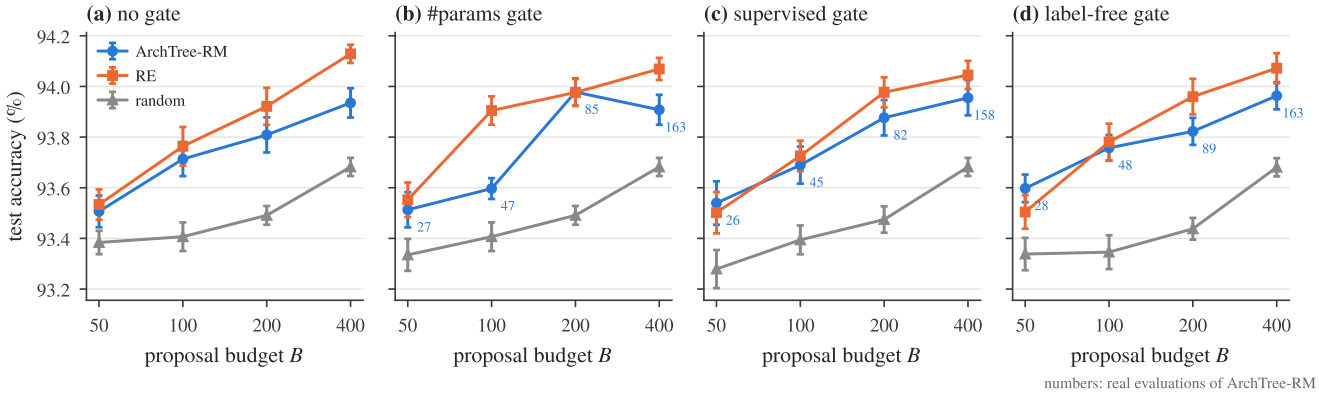


Figure S12. NAS-Bench-101: test accuracy (mean ± standard error) against the proposal budget for three proposers under each gate. Numbers give the real evaluations of ArchTree-RM.

Table S33. LLM prior strength: mean test accuracy for each α and, in brackets, the paired difference to $\alpha = 0$ (pt). No difference is significant after Holm correction.

α	CIFAR-10		CIFAR-100		ImageNet16-120	
	$B = 50$	$B = 100$	$B = 50$	$B = 100$	$B = 50$	$B = 100$
0 (off)	94.04	94.19	72.19	73.02	45.91	46.45
0.05	94.21 (+0.17)	94.36 (+0.17)	72.93 (+0.74)	73.21 (+0.19)	45.77 (−0.15)	46.34 (−0.11)
0.1	94.15 (+0.11)	94.34 (+0.16)	72.69 (+0.51)	73.14 (+0.12)	46.00 (+0.09)	46.21 (−0.25)
0.15	94.25 (+0.21)	94.27 (+0.09)	72.96 (+0.77)	73.20 (+0.18)	45.69 (−0.22)	45.82 (−0.63)
0.2	94.24 (+0.21)	94.29 (+0.11)	73.04 (+0.85)	73.31 (+0.29)	45.87 (−0.05)	46.44 (−0.02)
0.3	94.24 (+0.21)	94.29 (+0.11)	73.04 (+0.85)	73.31 (+0.29)	45.87 (−0.05)	46.44 (−0.02)
0.5	94.24 (+0.21)	94.29 (+0.11)	73.04 (+0.85)	73.31 (+0.29)	45.87 (−0.05)	46.44 (−0.02)
0.7	94.24 (+0.21)	94.29 (+0.11)	73.04 (+0.85)	73.31 (+0.29)	45.87 (−0.05)	46.44 (−0.02)
1	94.24 (+0.21)	94.29 (+0.11)	73.04 (+0.85)	73.31 (+0.29)	45.87 (−0.05)	46.44 (−0.02)

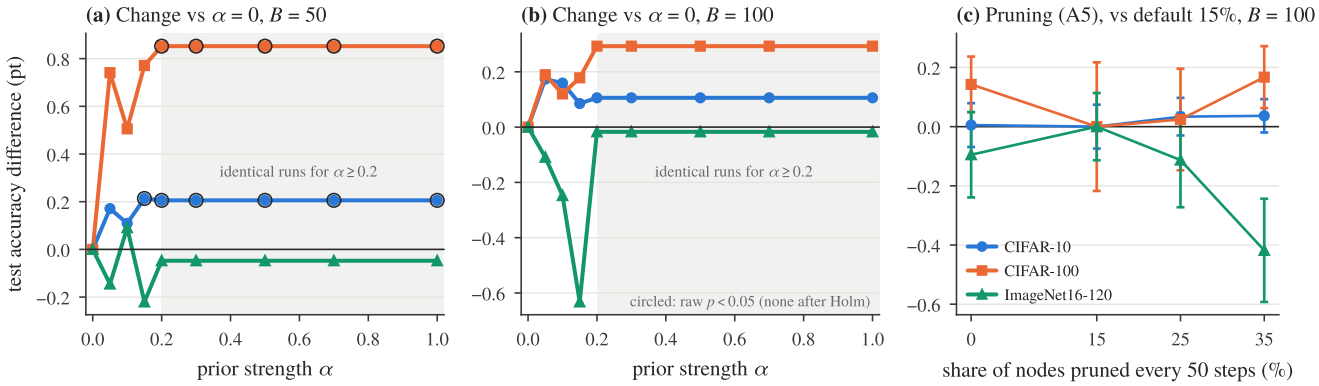


Figure S13. (a), (b) Change in test accuracy relative to $\alpha = 0$ for each prior strength at budgets 50 and 100 (circled: $p < 0.05$ before correction). The shaded region gives identical runs. (c) Effect of the pruned share (A5) relative to the default of 15% (mean ± standard error).

Table S34. Convergence at $B = 100$ (means over 12 seeds): best validation accuracy after 10 and 25 real evaluations, final value, and real evaluations needed to come within 2, 1 and 0.5 pt of the best validation accuracy in the space (seeds that never got there are counted at the end of their run; in brackets: seeds that got there, if fewer than 12).

Dataset	Arm	best@10	best@25	final	evals to 2 pt	to 1 pt	to 0.5 pt	real evals
CIFAR-10	prior off (α=0)	90.38	91.19	91.47	5.0	15.5	22.9	47.8
	LLM prior (standard)	91.03	91.47	91.55	2.0	5.0	12.5	45.0
	LLM prior (described ops)	90.74	91.37	91.53	1.0	7.8	23.3	45.8
	size prior (no LLM)	90.20	91.52	91.56	1.0	16.4	16.4	37.6
CIFAR-100	prior off (α=0)	70.21	72.10	73.10	22.0 (11/12)	30.8 (11/12)	36.1 (8/12)	47.1
	LLM prior (standard)	70.75	73.08	73.30	13.2	15.6	27.1 (9/12)	42.8
	LLM prior (described ops)	70.81	72.40	73.20	21.2	25.4	35.7 (9/12)	43.6
	size prior (no LLM)	70.83	73.15	73.43	17.4	19.7	20.6	41.0
ImageNet16-120	prior off (α=0)	43.75	45.84	46.42	14.2	17.8	24.4 (11/12)	45.4
	LLM prior (standard)	43.90	45.96	46.49	14.6	23.2	32.6 (11/12)	48.2
	LLM prior (described ops)	44.98	46.07	46.47	1.0	19.4	26.2	44.8
	size prior (no LLM)	42.80	45.73	46.08	15.4	31.2 (8/12)	34.7 (6/12)	46.4

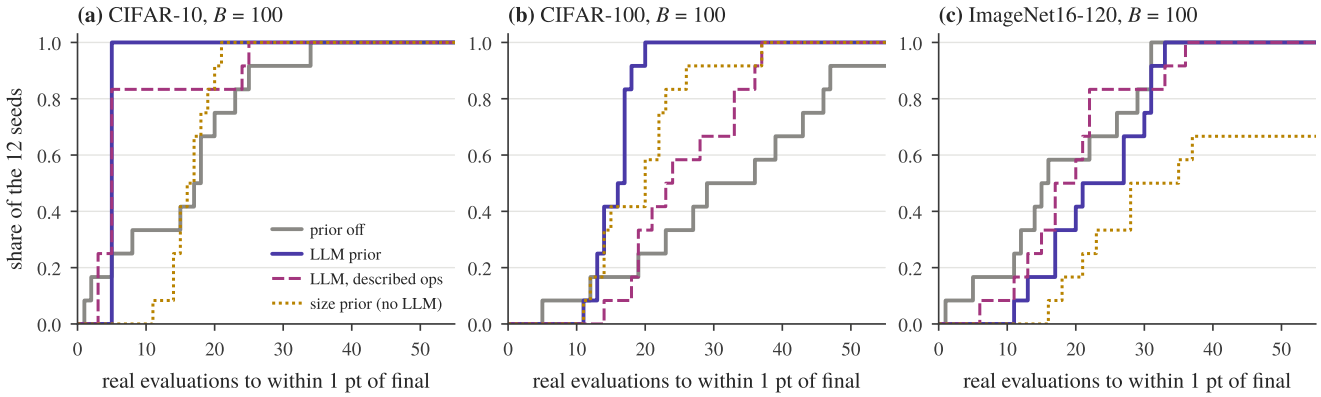


Figure S14. Share of the 12 seeds that have come within 1 pt of the best validation accuracy in the space after a given number of real evaluations ($B = 100$). Curves that end below 1 have seeds that never got there.

on CIFAR-100 15.6 (30.8, with one seed never getting there); on ImageNet16-120 it is slower (23.2 against 17.8). The prior with described operations is also fast on CIFAR-10 (7.8), and the size prior without any LLM matches much of the gain on CIFAR-100 (19.7) but not on ImageNet16-120. Final validation accuracies of the four arms are within 0.5 pt of each other on every dataset.

What the LLM knows. Tables S35 and S36 and Figure S15 test whether the prior reflects knowledge of operations or recall of the benchmark. With operation names the LLM ranks the five operations with $\rho = 0.50$–0.57; with the operations described but not named it does better (0.64–0.70), and with anonymous tokens it falls to 0.14–0.19, which is still above chance. A prior that simply prefers larger operations, with no LLM, scores higher than every LLM variant (0.73–0.76). The LLM therefore encodes mostly general knowledge about operation capacity, not memorised benchmark tables. Table S37 shows that a 7B model ranks worse (0.41) and fails to answer in 3.8% of calls, whereas the 20B and 120B models are similar in ρ (0.60 and 0.61), although the larger one more often puts the best operation first (0.65 against 0.46).

S11 Hardware-aware search

Table S38 gives hypervolume (relative to random search), inverted generational distance (IGD) and front size for every device and budget, and Figure S16 the IGD and front-size curves. The hypervolumes of ArchTree-HW and NSGA-II-style search lie within 0.5% of random search everywhere (99.89–100.42%), and all 96 tests are ties after Holm correction (22 have $p < 0.05$ before it, mostly small hypervolume gains of ArchTree-HW): on HW-NAS-Bench the accuracy–latency front is easy to cover, and no difference between the methods survives correction. IGD separates them somewhat more: at 25 proposals ArchTree-HW is the furthest from the true front on all six devices, and NSGA-II-style search finds the largest fronts in 18 of 24 device–budget cells (Figure S16b); at 200 proposals the differences are small, NSGA-II-style search having the lowest IGD on four of the six devices.

Table S35. LLM knowledge probe on NAS-Bench-201 (56 prefixes per dataset): mean Spearman ρ between the scores given to the candidate operations and their true ranking, and share of prefixes where the top-scored operation is the best.

Probe	CIFAR-10		CIFAR-100		ImageNet16-120	
	ρ	top-1	ρ	top-1	ρ	top-1
standard prompt (op names)	0.57	0.52	0.50	0.55	0.52	0.45
operations described, names hidden	0.70	0.57	0.64	0.52	0.66	0.54
anonymous tokens	0.14	0.11	0.19	0.11	0.18	0.18
size prior (no LLM)	0.76	0.80	0.75	0.86	0.73	0.73

Table S36. Paired tests of the probe (Wilcoxon over the 56 prefixes, Holm-corrected; difference in Spearman ρ).

Dataset	Comparison	n	mean diff. (ρ)	p_{Holm}	sig.
CIFAR-10	anonymous vs chance	56	0.140	0.029	yes
	described vs chance	56	0.701	<0.001	yes
	size prior vs chance	56	0.765	<0.001	yes
	standard vs chance	56	0.574	<0.001	yes
	standard vs anonymous	56	0.434	<0.001	yes
	standard vs described	56	−0.127	0.002	yes
	standard vs size prior	56	−0.191	0.001	yes
CIFAR-100	anonymous vs chance	56	0.190	0.007	yes
	described vs chance	56	0.640	<0.001	yes
	size prior vs chance	56	0.751	<0.001	yes
	standard vs chance	56	0.500	<0.001	yes
	standard vs anonymous	56	0.310	<0.001	yes
	standard vs described	56	−0.141	0.001	yes
	standard vs size prior	56	−0.251	<0.001	yes
ImageNet16-120	anonymous vs chance	56	0.181	0.007	yes
	described vs chance	56	0.661	<0.001	yes
	size prior vs chance	56	0.733	<0.001	yes
	standard vs chance	56	0.516	<0.001	yes
	standard vs anonymous	56	0.334	<0.001	yes
	standard vs described	56	−0.145	<0.001	yes
	standard vs size prior	56	−0.217	<0.001	yes

Table S37. Model size (CIFAR-10, 26 prefixes): mean Spearman ρ, top-1 hit rate, share of calls that fell back to uniform scores and operation-name errors per call.

Model	prefixes	mean ρ	top-1 hit	fallback (%)	op-name errors / call
`allam-2-7b`	26	0.41	0.50	3.8	0.50
`openai/gpt-oss-120b`	26	0.61	0.65	0.0	0.00
`openai/gpt-oss-20b`	26	0.60	0.46	0.0	0.00

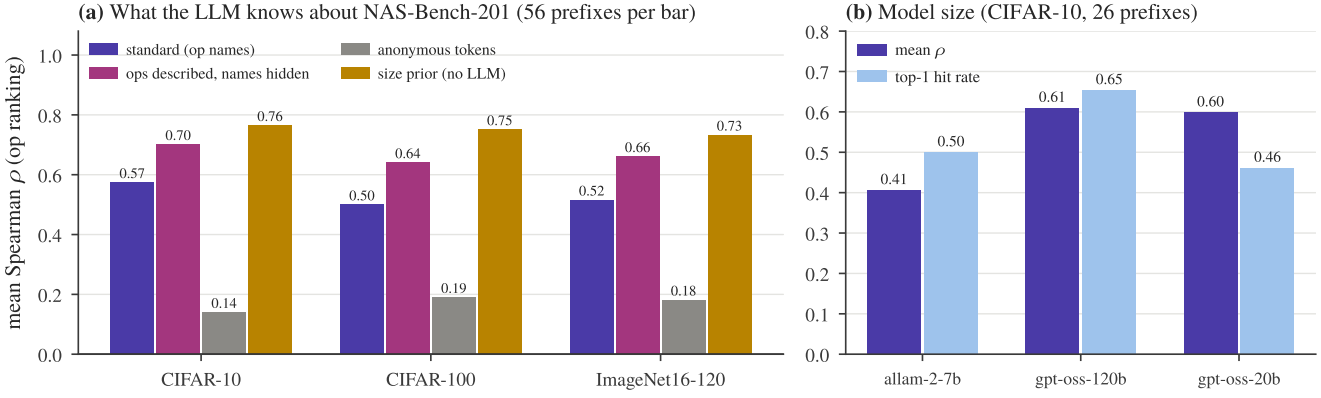


Figure S15. (a) Mean Spearman correlation between LLM scores and the true ranking of operations under four prompt conditions. (b) The same for three model sizes on CIFAR-10.

Table S38. HW-NAS-Bench (CIFAR-10): hypervolume as % of random search, IGD to the true front, and mean size of the front found by ArchTree-HW (12 seeds).

Device	B	HV (% of random)		IGD			front
		ArchTree-HW	NSGA-II-style	ArchTree-HW	NSGA-II-style	random	size
Edge GPU	25	100.04	99.96	23.31	14.22	13.67	5.2
	50	100.38	100.15	15.98	9.94	9.81	6.4
	100	100.39	100.09	7.03	2.52	7.21	9.7
	200	100.32	100.13	0.22	0.17	0.38	12.6
Raspberry Pi 4	25	99.89	99.95	11.07	9.24	8.08	6.6
	50	100.23	100.07	7.73	8.04	5.90	10.1
	100	100.05	100.19	2.60	4.06	4.35	12.9
	200	100.14	100.31	2.04	1.78	1.31	18.4
Edge TPU	25	100.01	99.94	1.35	1.16	1.04	3.8
	50	100.42	100.24	0.65	0.80	0.76	7.9
	100	100.23	100.15	0.34	0.48	0.50	10.8
	200	100.24	100.27	0.22	0.15	0.26	13.3
Pixel 3	25	100.08	99.96	9.36	7.98	6.92	7.0
	50	100.18	100.02	7.71	7.41	4.97	9.4
	100	100.27	100.06	4.40	6.12	3.73	12.6
	200	100.21	100.22	0.69	1.44	0.80	18.0
Eyeriss	25	100.16	99.94	1.54	1.28	1.24	3.8
	50	100.35	100.09	1.17	0.99	1.07	6.5
	100	100.32	100.25	0.78	0.58	0.81	8.7
	200	100.34	100.26	0.51	0.34	0.64	11.0
FPGA	25	100.25	99.96	1.33	1.15	1.15	5.8
	50	100.37	99.98	1.06	0.95	1.02	7.4
	100	100.36	100.08	0.78	0.60	0.82	8.7
	200	100.37	100.26	0.55	0.35	0.64	11.1

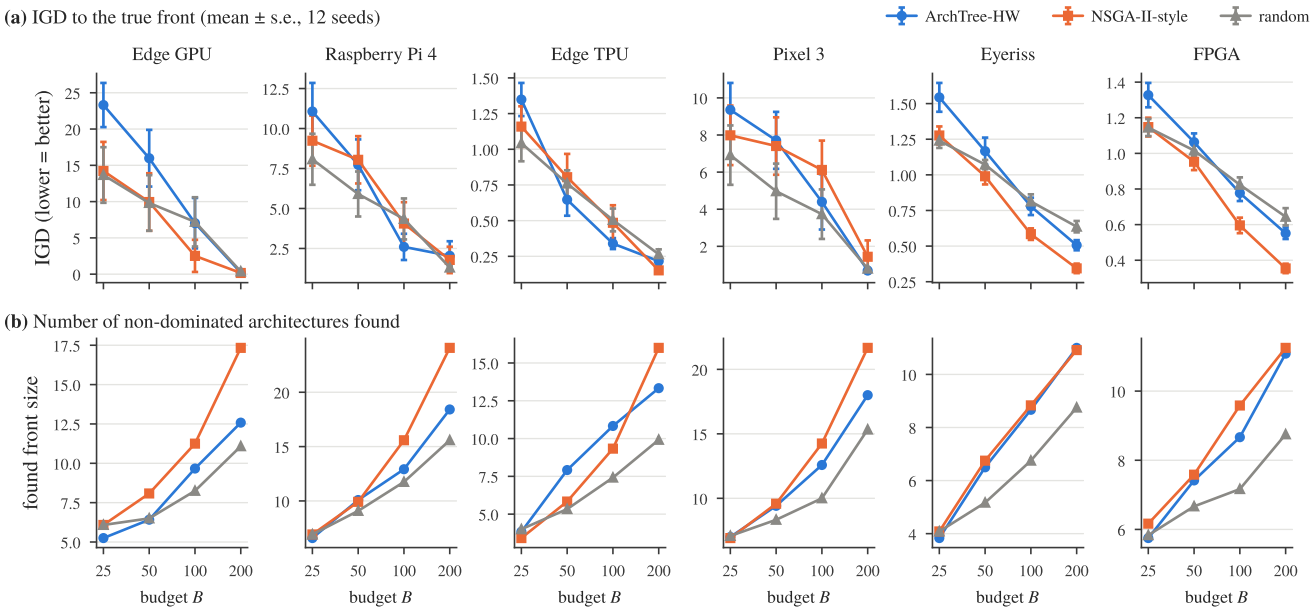


Figure S16. Hardware-aware search on the six HW-NAS-Bench devices. (a) IGD between the found and the true accuracy–latency front (mean ± standard error). (b) Number of non-dominated architectures found.

Table S39. True Pareto fronts of each device pair (CIFAR-10): front sizes, shared architectures, Jaccard index, Spearman correlation of latency over all architectures, and hypervolume ratio when the front of one device is used on the other.

Device pair (a – b)	front sizes	shared	Jaccard	ρ(latency)	HV ratio $a \to b$	HV ratio $b \to a$
Edge GPU – Raspberry Pi 4	24 / 48	6	0.09	0.65	0.992	0.998
Edge GPU – Edge TPU	24 / 43	4	0.06	0.70	0.991	0.989
Edge GPU – Pixel 3	24 / 55	5	0.07	0.63	0.988	0.997
Edge GPU – Eyeriss	24 / 17	3	0.08	0.51	0.969	0.833
Edge GPU – FPGA	24 / 11	4	0.13	0.62	0.950	0.833
Raspberry Pi 4 – Edge TPU	48 / 43	9	0.11	0.59	0.996	0.995
Raspberry Pi 4 – Pixel 3	48 / 55	13	0.14	0.97	0.999	0.998
Raspberry Pi 4 – Eyeriss	48 / 17	7	0.12	0.68	1.000	0.994
Raspberry Pi 4 – FPGA	48 / 11	7	0.13	0.89	1.000	0.994
Edge TPU – Pixel 3	43 / 55	8	0.09	0.63	0.987	0.996
Edge TPU – Eyeriss	43 / 17	6	0.11	0.82	0.995	0.988
Edge TPU – FPGA	43 / 11	7	0.15	0.53	0.998	0.988
Pixel 3 – Eyeriss	55 / 17	9	0.14	0.72	0.999	0.986
Pixel 3 – FPGA	55 / 11	7	0.12	0.87	0.998	0.985
Eyeriss – FPGA	17 / 11	8	0.40	0.75	1.000	0.999

Table S40. Reproducibility checks carried out on the final code.

Check	Result
Parallel workers (equal GPU-hours)	1 and 3 workers give identical results (252 of 252 runs)
Device rerun against the offline replica	identical (Phase 3: 108 of 108 runs; design ablation: 540 of 540)
A1, A3, A6 against the headline runs	identical (144 of 144 runs each; Table S26)
Zero-information LLM prior against $\alpha = 0$	identical on 24 of 24 seeds
Repeated real evaluations in ArchTree runs	0 in every run (Table S11)
NAS-Bench-201 path after the NB101 refactor	bit-identical (6 of 6 regression runs)
TensorFlow-free NAS-Bench-101 reader	all 423,624 graph hashes reproduced
Shared proxy cache under a file lock	no entry lost with 4 concurrent workers
HW-NAS-Bench architecture alignment	accuracy–latency Spearman $\rho = 0.624$ after re-indexing
Equal-GPU-hour runs below budget	5 of 1,128 (proposals exhausted at $k = 100$; section S5)
LLM fallbacks	0 in LLM-prior runs; 0.37 random replacements per LLM-NAS run

Table S41. Data files used for the paper (SHA-256).

File	Size (bytes)	SHA-256
NAS-Bench-201-v1_1-096897.pth	5,003,180,601	`2317a7dc911654322f4f932730a821cf38ecc5b5f1adde0aff26ff6b3f736666`
nasbench_only108.tfrecord	522,767,376	`4c39c3936e36a85269881d659e44e61a245babcb72cb374eacacf75d0e5f4fd1`
nb201_cache_v2.json	4,894,199	`b6f19d2d0adee7b09d3a9ff098a379695d3c224cdcf5dafeb25f0924afee2694`
nb101_cache.npz	22,126,761	`4e0fb9901ff6952fc1d345c6ee1be8f6c7b4406c0f0843bbd255bc14a302d19e`
natssss_cache.json	12,828,328	`8cac2574ab56b07dcf9b82461e03d0195a34a0187eb4239b362f9400cdd28d18a`
HW-NAS-Bench-v1_0.pickle	10,782,467	`57f92c2d7b88814fa337d00f4597c8ca30d507c5a855c18d6fac57e5ad872700`

Table S39 compares the true fronts of the devices. Few architectures are shared (Jaccard 0.06–0.40) although latencies are correlated (ρ = 0.51–0.97). Yet a front computed for one device, when evaluated on another, keeps 83–100% of the hypervolume of that device's true front. Fronts differ in membership far more than in quality, so a search for one device is a reasonable starting point for another, while the exact Pareto set must be searched per device.

S12 Seeds, reproducibility checks and data

Seeds. Main seeds (all experiments): 42, 111, 123, 222, 333, 456, 789, 999, 1001, 2024, 31415, 27182. Held-out seeds (design ablation only, never used for a design decision): 7, 13, 77, 101, 202, 303, 404, 505, 606, 707, 808, 909. Proxy calibration draws its 1,500 architectures with seed 0, and the proxy network is initialised with seed 1234. Every run seeds its own random number generators, so results do not depend on the order or the number of parallel workers.

Run order. The results are regenerated by building the benchmark caches (`tools/build_nb201_cache.py`, `tools/build_nb101_cache.py`), running the experiments in the order of Table 3 of the main paper (`experiments/run_phase0.py` to `run_equal_gpu_hours.py`), aggregating the statistics (`run_phase6_aggregate.py`) and finally running `make_figures.py` and the two supplementary scripts. The PowerShell script `run_full_pipeline.ps1` in the repository performs all steps in sequence.

Table S42. Software and hardware environment of the runs.

Item	Version	Item	Version
Python	3.10.9	PyTorch / torchvision	2.5.1 / 0.20.1 (CUDA 12.4)
NumPy	2.2.6	SciPy	1.15.3
pandas	2.3.3	Matplotlib	3.10.9
nas-bench-201	2.1	groq client	1.5.0
GPU	NVIDIA GeForce RTX 2050, 4 GB	OS	Windows 11 (build 26300)

The complete pinned package list is `outputs/final/repro/requirements.lock.txt` in the repository. Tabular look-ups are deterministic; the proxy kernels use cuDNN benchmark mode, so raw proxy scores are cached and shipped with the code to make them identical across machines. LLM answers are cached per prompt and shipped for exact replay.

References

[1] Kocsis L and Szepesvári C 2006 Bandit based Monte-Carlo planning. In: *Proceedings of the European Conference on Machine Learning*, LNCS 4212, pp. 282–293

[2] Coquelin P A and Munos R 2007 Bandit algorithms for tree search. In: *Proceedings of the 23rd Conference on Uncertainty in Artificial Intelligence*, pp. 67–74

[3] Bubeck S, Munos R, Stoltz G and Szepesvári C 2011 X-armed bandits. *J. Mach. Learn. Res.* 12: 1655–1695